

# FRM PART I – MOCK I ANSWERS

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*By AnalystPrep*

Reference Table: Let Z be a standard normal random variable.

| z     | P(Z<z) | z     | P(Z<z) | z     | P(Z<z) | z     | P(Z<z) | z     | P(Z<z) | z     | P(Z<z) |
|-------|--------|-------|--------|-------|--------|-------|--------|-------|--------|-------|--------|
| -3    | 0.0013 | -2.50 | 0.0062 | -2.00 | 0.0228 | -1.50 | 0.0668 | -1.00 | 0.1587 | -0.50 | 0.3085 |
| -2.99 | 0.0014 | -2.49 | 0.0064 | -1.99 | 0.0233 | -1.49 | 0.0681 | -0.99 | 0.1611 | -0.49 | 0.3121 |
| -2.98 | 0.0014 | -2.48 | 0.0066 | -1.98 | 0.0239 | -1.48 | 0.0694 | -0.98 | 0.1635 | -0.48 | 0.3156 |
| -2.97 | 0.0015 | -2.47 | 0.0068 | -1.97 | 0.0244 | -1.47 | 0.0708 | -0.97 | 0.1660 | -0.47 | 0.3192 |
| -2.96 | 0.0015 | -2.46 | 0.0069 | -1.96 | 0.0250 | -1.46 | 0.0721 | -0.96 | 0.1685 | -0.46 | 0.3228 |
| -2.95 | 0.0016 | -2.45 | 0.0071 | -1.95 | 0.0256 | -1.45 | 0.0735 | -0.95 | 0.1711 | -0.45 | 0.3264 |
| -2.94 | 0.0016 | -2.44 | 0.0073 | -1.94 | 0.0262 | -1.44 | 0.0749 | -0.94 | 0.1736 | -0.44 | 0.3300 |
| -2.93 | 0.0017 | -2.43 | 0.0075 | -1.93 | 0.0268 | -1.43 | 0.0764 | -0.93 | 0.1762 | -0.43 | 0.3336 |
| -2.92 | 0.0018 | -2.42 | 0.0078 | -1.92 | 0.0274 | -1.42 | 0.0778 | -0.92 | 0.1788 | -0.42 | 0.3372 |
| -2.91 | 0.0018 | -2.41 | 0.0080 | -1.91 | 0.0281 | -1.41 | 0.0793 | -0.91 | 0.1814 | -0.41 | 0.3409 |
| -2.9  | 0.0019 | -2.40 | 0.0082 | -1.90 | 0.0287 | -1.40 | 0.0808 | -0.90 | 0.1841 | -0.40 | 0.3446 |
| -2.89 | 0.0019 | -2.39 | 0.0084 | -1.89 | 0.0294 | -1.39 | 0.0823 | -0.89 | 0.1867 | -0.39 | 0.3483 |
| -2.88 | 0.0020 | -2.38 | 0.0087 | -1.88 | 0.0301 | -1.38 | 0.0838 | -0.88 | 0.1894 | -0.38 | 0.3520 |
| -2.87 | 0.0021 | -2.37 | 0.0089 | -1.87 | 0.0307 | -1.37 | 0.0853 | -0.87 | 0.1922 | -0.37 | 0.3557 |
| -2.86 | 0.0021 | -2.36 | 0.0091 | -1.86 | 0.0314 | -1.36 | 0.0869 | -0.86 | 0.1949 | -0.36 | 0.3594 |
| -2.85 | 0.0022 | -2.35 | 0.0094 | -1.85 | 0.0322 | -1.35 | 0.0885 | -0.85 | 0.1977 | -0.35 | 0.3632 |
| -2.84 | 0.0023 | -2.34 | 0.0096 | -1.84 | 0.0329 | -1.34 | 0.0901 | -0.84 | 0.2005 | -0.34 | 0.3669 |
| -2.83 | 0.0023 | -2.33 | 0.0099 | -1.83 | 0.0336 | -1.33 | 0.0918 | -0.83 | 0.2033 | -0.33 | 0.3707 |
| -2.82 | 0.0024 | -2.32 | 0.0102 | -1.82 | 0.0344 | -1.32 | 0.0934 | -0.82 | 0.2061 | -0.32 | 0.3745 |
| -2.81 | 0.0025 | -2.31 | 0.0104 | -1.81 | 0.0351 | -1.31 | 0.0951 | -0.81 | 0.2090 | -0.31 | 0.3783 |
| -2.8  | 0.0026 | -2.30 | 0.0107 | -1.80 | 0.0359 | -1.30 | 0.0968 | -0.80 | 0.2119 | -0.30 | 0.3821 |
| -2.79 | 0.0026 | -2.29 | 0.0110 | -1.79 | 0.0367 | -1.29 | 0.0985 | -0.79 | 0.2148 | -0.29 | 0.3859 |
| -2.78 | 0.0027 | -2.28 | 0.0113 | -1.78 | 0.0375 | -1.28 | 0.1003 | -0.78 | 0.2177 | -0.28 | 0.3897 |
| -2.77 | 0.0028 | -2.27 | 0.0116 | -1.77 | 0.0384 | -1.27 | 0.1020 | -0.77 | 0.2206 | -0.27 | 0.3936 |
| -2.76 | 0.0029 | -2.26 | 0.0119 | -1.76 | 0.0392 | -1.26 | 0.1038 | -0.76 | 0.2236 | -0.26 | 0.3974 |
| -2.75 | 0.0030 | -2.25 | 0.0122 | -1.75 | 0.0401 | -1.25 | 0.1056 | -0.75 | 0.2266 | -0.25 | 0.4013 |
| -2.74 | 0.0031 | -2.24 | 0.0125 | -1.74 | 0.0409 | -1.24 | 0.1075 | -0.74 | 0.2296 | -0.24 | 0.4052 |
| -2.73 | 0.0032 | -2.23 | 0.0129 | -1.73 | 0.0418 | -1.23 | 0.1093 | -0.73 | 0.2327 | -0.23 | 0.4090 |
| -2.72 | 0.0033 | -2.22 | 0.0132 | -1.72 | 0.0427 | -1.22 | 0.1112 | -0.72 | 0.2358 | -0.22 | 0.4129 |
| -2.71 | 0.0034 | -2.21 | 0.0136 | -1.71 | 0.0436 | -1.21 | 0.1131 | -0.71 | 0.2389 | -0.21 | 0.4168 |
| -2.7  | 0.0035 | -2.20 | 0.0139 | -1.70 | 0.0446 | -1.20 | 0.1151 | -0.70 | 0.2420 | -0.20 | 0.4207 |
| -2.69 | 0.0036 | -2.19 | 0.0143 | -1.69 | 0.0455 | -1.19 | 0.1170 | -0.69 | 0.2451 | -0.19 | 0.4247 |
| -2.68 | 0.0037 | -2.18 | 0.0146 | -1.68 | 0.0465 | -1.18 | 0.1190 | -0.68 | 0.2483 | -0.18 | 0.4286 |
| -2.67 | 0.0038 | -2.17 | 0.0150 | -1.67 | 0.0475 | -1.17 | 0.1210 | -0.67 | 0.2514 | -0.17 | 0.4325 |
| -2.66 | 0.0039 | -2.16 | 0.0154 | -1.66 | 0.0485 | -1.16 | 0.1230 | -0.66 | 0.2546 | -0.16 | 0.4364 |
| -2.65 | 0.0040 | -2.15 | 0.0158 | -1.65 | 0.0495 | -1.15 | 0.1251 | -0.65 | 0.2578 | -0.15 | 0.4404 |
| -2.64 | 0.0041 | -2.14 | 0.0162 | -1.64 | 0.0505 | -1.14 | 0.1271 | -0.64 | 0.2611 | -0.14 | 0.4443 |
| -2.63 | 0.0043 | -2.13 | 0.0166 | -1.63 | 0.0516 | -1.13 | 0.1292 | -0.63 | 0.2643 | -0.13 | 0.4483 |
| -2.62 | 0.0044 | -2.12 | 0.0170 | -1.62 | 0.0526 | -1.12 | 0.1314 | -0.62 | 0.2676 | -0.12 | 0.4522 |
| -2.61 | 0.0045 | -2.11 | 0.0174 | -1.61 | 0.0537 | -1.11 | 0.1335 | -0.61 | 0.2709 | -0.11 | 0.4562 |
| -2.6  | 0.0047 | -2.10 | 0.0179 | -1.60 | 0.0548 | -1.10 | 0.1357 | -0.60 | 0.2743 | -0.10 | 0.4602 |
| -2.59 | 0.0048 | -2.09 | 0.0183 | -1.59 | 0.0559 | -1.09 | 0.1379 | -0.59 | 0.2776 | -0.09 | 0.4641 |
| -2.58 | 0.0049 | -2.08 | 0.0188 | -1.58 | 0.0571 | -1.08 | 0.1401 | -0.58 | 0.2810 | -0.08 | 0.4681 |
| -2.57 | 0.0051 | -2.07 | 0.0192 | -1.57 | 0.0582 | -1.07 | 0.1423 | -0.57 | 0.2843 | -0.07 | 0.4721 |
| -2.56 | 0.0052 | -2.06 | 0.0197 | -1.56 | 0.0594 | -1.06 | 0.1446 | -0.56 | 0.2877 | -0.06 | 0.4761 |
| -2.55 | 0.0054 | -2.05 | 0.0202 | -1.55 | 0.0606 | -1.05 | 0.1469 | -0.55 | 0.2912 | -0.05 | 0.4801 |
| -2.54 | 0.0055 | -2.04 | 0.0207 | -1.54 | 0.0618 | -1.04 | 0.1492 | -0.54 | 0.2946 | -0.04 | 0.4840 |
| -2.53 | 0.0057 | -2.03 | 0.0212 | -1.53 | 0.0630 | -1.03 | 0.1515 | -0.53 | 0.2981 | -0.03 | 0.4880 |
| -2.52 | 0.0059 | -2.02 | 0.0217 | -1.52 | 0.0643 | -1.02 | 0.1539 | -0.52 | 0.3015 | -0.02 | 0.4920 |
| -2.51 | 0.0060 | -2.01 | 0.0222 | -1.51 | 0.0655 | -1.01 | 0.1562 | -0.51 | 0.3050 | -0.01 | 0.4960 |

Reference Table: t-table.

| cum. prob | $t_{.50}$               | $t_{.75}$ | $t_{.80}$ | $t_{.85}$ | $t_{.90}$ | $t_{.95}$ | $t_{.975}$ | $t_{.99}$ | $t_{.995}$ | $t_{.999}$ | $t_{.9995}$ |
|-----------|-------------------------|-----------|-----------|-----------|-----------|-----------|------------|-----------|------------|------------|-------------|
| one-tail  | 0.50                    | 0.25      | 0.20      | 0.15      | 0.10      | 0.05      | 0.025      | 0.01      | 0.005      | 0.001      | 0.0005      |
| two-tails | 1.00                    | 0.50      | 0.40      | 0.30      | 0.20      | 0.10      | 0.05       | 0.02      | 0.01       | 0.002      | 0.001       |
| df        |                         |           |           |           |           |           |            |           |            |            |             |
| 1         | 0.000                   | 1.000     | 1.376     | 1.963     | 3.078     | 6.314     | 12.71      | 31.82     | 63.66      | 318.31     | 636.62      |
| 2         | 0.000                   | 0.816     | 1.061     | 1.386     | 1.886     | 2.920     | 4.303      | 6.965     | 9.925      | 22.327     | 31.599      |
| 3         | 0.000                   | 0.765     | 0.978     | 1.250     | 1.638     | 2.353     | 3.182      | 4.541     | 5.841      | 10.215     | 12.924      |
| 4         | 0.000                   | 0.741     | 0.941     | 1.190     | 1.533     | 2.132     | 2.776      | 3.747     | 4.604      | 7.173      | 8.610       |
| 5         | 0.000                   | 0.727     | 0.920     | 1.156     | 1.476     | 2.015     | 2.571      | 3.365     | 4.032      | 5.893      | 6.869       |
| 6         | 0.000                   | 0.718     | 0.906     | 1.134     | 1.440     | 1.943     | 2.447      | 3.143     | 3.707      | 5.208      | 5.959       |
| 7         | 0.000                   | 0.711     | 0.896     | 1.119     | 1.415     | 1.895     | 2.365      | 2.998     | 3.499      | 4.785      | 5.408       |
| 8         | 0.000                   | 0.706     | 0.889     | 1.108     | 1.397     | 1.860     | 2.306      | 2.896     | 3.355      | 4.501      | 5.041       |
| 9         | 0.000                   | 0.703     | 0.883     | 1.100     | 1.383     | 1.833     | 2.262      | 2.821     | 3.250      | 4.297      | 4.781       |
| 10        | 0.000                   | 0.700     | 0.879     | 1.093     | 1.372     | 1.812     | 2.228      | 2.764     | 3.169      | 4.144      | 4.587       |
| 11        | 0.000                   | 0.697     | 0.876     | 1.088     | 1.363     | 1.796     | 2.201      | 2.718     | 3.106      | 4.025      | 4.437       |
| 12        | 0.000                   | 0.695     | 0.873     | 1.083     | 1.356     | 1.782     | 2.179      | 2.681     | 3.055      | 3.930      | 4.318       |
| 13        | 0.000                   | 0.694     | 0.870     | 1.079     | 1.350     | 1.771     | 2.160      | 2.650     | 3.012      | 3.852      | 4.221       |
| 14        | 0.000                   | 0.692     | 0.868     | 1.076     | 1.345     | 1.761     | 2.145      | 2.624     | 2.977      | 3.787      | 4.140       |
| 15        | 0.000                   | 0.691     | 0.866     | 1.074     | 1.341     | 1.753     | 2.131      | 2.602     | 2.947      | 3.733      | 4.073       |
| 16        | 0.000                   | 0.690     | 0.865     | 1.071     | 1.337     | 1.746     | 2.120      | 2.583     | 2.921      | 3.686      | 4.015       |
| 17        | 0.000                   | 0.689     | 0.863     | 1.069     | 1.333     | 1.740     | 2.110      | 2.567     | 2.898      | 3.646      | 3.965       |
| 18        | 0.000                   | 0.688     | 0.862     | 1.067     | 1.330     | 1.734     | 2.101      | 2.552     | 2.878      | 3.610      | 3.922       |
| 19        | 0.000                   | 0.688     | 0.861     | 1.066     | 1.328     | 1.729     | 2.093      | 2.539     | 2.861      | 3.579      | 3.883       |
| 20        | 0.000                   | 0.687     | 0.860     | 1.064     | 1.325     | 1.725     | 2.086      | 2.528     | 2.845      | 3.552      | 3.850       |
| 21        | 0.000                   | 0.686     | 0.859     | 1.063     | 1.323     | 1.721     | 2.080      | 2.518     | 2.831      | 3.527      | 3.819       |
| 22        | 0.000                   | 0.686     | 0.858     | 1.061     | 1.321     | 1.717     | 2.074      | 2.508     | 2.819      | 3.505      | 3.792       |
| 23        | 0.000                   | 0.685     | 0.858     | 1.060     | 1.319     | 1.714     | 2.069      | 2.500     | 2.807      | 3.485      | 3.768       |
| 24        | 0.000                   | 0.685     | 0.857     | 1.059     | 1.318     | 1.711     | 2.064      | 2.492     | 2.797      | 3.467      | 3.745       |
| 25        | 0.000                   | 0.684     | 0.856     | 1.058     | 1.316     | 1.708     | 2.060      | 2.485     | 2.787      | 3.450      | 3.725       |
| 26        | 0.000                   | 0.684     | 0.856     | 1.058     | 1.315     | 1.706     | 2.056      | 2.479     | 2.779      | 3.435      | 3.707       |
| 27        | 0.000                   | 0.684     | 0.855     | 1.057     | 1.314     | 1.703     | 2.052      | 2.473     | 2.771      | 3.421      | 3.690       |
| 28        | 0.000                   | 0.683     | 0.855     | 1.056     | 1.313     | 1.701     | 2.048      | 2.467     | 2.763      | 3.408      | 3.674       |
| 29        | 0.000                   | 0.683     | 0.854     | 1.055     | 1.311     | 1.699     | 2.045      | 2.462     | 2.756      | 3.396      | 3.659       |
| 30        | 0.000                   | 0.683     | 0.854     | 1.055     | 1.310     | 1.697     | 2.042      | 2.457     | 2.750      | 3.385      | 3.646       |
| 40        | 0.000                   | 0.681     | 0.851     | 1.050     | 1.303     | 1.684     | 2.021      | 2.423     | 2.704      | 3.307      | 3.551       |
| 60        | 0.000                   | 0.679     | 0.848     | 1.045     | 1.296     | 1.671     | 2.000      | 2.390     | 2.660      | 3.232      | 3.460       |
| 80        | 0.000                   | 0.678     | 0.846     | 1.043     | 1.292     | 1.664     | 1.990      | 2.374     | 2.639      | 3.195      | 3.416       |
| 100       | 0.000                   | 0.677     | 0.845     | 1.042     | 1.290     | 1.660     | 1.984      | 2.364     | 2.626      | 3.174      | 3.390       |
| 1000      | 0.000                   | 0.675     | 0.842     | 1.037     | 1.282     | 1.646     | 1.962      | 2.330     | 2.581      | 3.098      | 3.300       |
| <b>Z</b>  | 0.000                   | 0.674     | 0.842     | 1.036     | 1.282     | 1.645     | 1.960      | 2.326     | 2.576      | 3.090      | 3.291       |
|           | 0%                      | 50%       | 60%       | 70%       | 80%       | 90%       | 95%        | 98%       | 99%        | 99.8%      | 99.9%       |
|           | <b>Confidence Level</b> |           |           |           |           |           |            |           |            |            |             |

## **Foundations of Risk Management**

1. Hannah Zane is a fixed-income portfolio manager at Smart Capital. She comes across a research report issued by a major brokerage firm on Fortunes Corporation in which the research analyst argues that within one year, the Fortune's credit fundamentals will strengthen. The report goes further to assert following the narrowing spread, the market will respond by demanding a lower credit spread. After reading the report, Ms. Zane decides that she wants to take a credit view on Fortunes Corporation.

Next week, Fortunes Corporation has declared its intention to bring to the market a 10-year senior bond issue at par with a coupon rate of 12%, offering a spread of 800 basis points over the corresponding 10-year Treasury issue. Ms. Zane is not keen to purchase the bond outright because she does not want to bear the out-of-pocket costs, the inconvenience of arranging the financing, actually going long the bond and taking delivery. Instead, she would like to express her view on Fortune Corporation's credit risk by entering into a total return swap that matures in one year with the senior bonds that are about to be issued as the reference obligation. Under the terms of the contract, payments will be exchanged semiannually, where the total return receiver will pay the six-month Treasury rate plus 350 basis points.

Which of the following is most likely correct?

- A. Ms. Zane's counterparty will enter the total return swap as the total return receiver.
- B. Ms. Zane will enter the total return swap as the total return receiver.
- C. The total return receiver will be exposed to credit risk but not interest rate risk.
- D. The party that agrees to receive the floating payments and pay the total return is the total return receiver.

Correct answer: B.

Explanation: In a total return swap, one party, the payer, is able to confidently remove all the economic exposure of the asset without having to sell it. The receiver of a total return swap, on the other hand, is able to access the economic exposure of the asset without having to buy it.

The party that agrees to make the floating payments and receive the total return is the total return receiver, while the party that agrees to receive the floating payments and pay the total return is the total return payer. The total return receiver is exposed not only to credit risk but also to interest-rate risk. If the credit risk spread declines, for example, there would be favorable price movement (price increase) for the reference obligation, but any gain made on that front can be offset by a rise in the level of interest rates.

Expecting the credit risk spread to decline (in line with the report) and not expecting interest rates to decline, Ms. Zane would enter the total return swap as the total return receiver. Based on the report, she has reasons to believe that the fortunes of Fortune Corporation will improve over the next year, triggering a decline in the firm's credit spread relative to U.S. Treasury securities, which will, in turn, result in a favorable price movement. As the total return receiver, Ms. Zane would profit from the decrease in spread.

***FRM Part 1, Book 1: Foundations of Risk Management, Chapter 4, Credit Risk Transfer Mechanisms, Learning objective: Compare different types of credit derivatives, explain how each one transfers credit risk, and describe their advantages and disadvantages.***

2. The macroeconomic research department of a large investment bank uses the arbitrage pricing theory to set up its expectations on equity market performance in the next quarter. It uses the two-factor model based on the short-term interest rate level and the GDP growth rate. The risk-free rate is currently 1%.

|                          | Factor Beta | Factor Risk Premium |
|--------------------------|-------------|---------------------|
| Short-term interest rate | -0.9        | 3%                  |
| GDP                      | 1.3         | 7%                  |

Although market participants do not expect any changes in interest rate, analysts at the bank are strongly confident that there will be an increase of 100 basis points. They also believe the GDP growth rate will beat market expectations by 50 basis points.

Taking into account the expectations of the macroeconomic research department, what is the expected return on the equity market?

- A. 7.15%
- B. 7.25%
- C. 7.40%
- D. 8.30%

Correct answer: A

Explanation: Currently, the market expects returns on equity markets of  $1\% + (-0.9) \times 3\% + 1.3 \times 7\% = 7.4\%$ .

Expected return, assuming the surprises proposed by the macroeconomic research department, will equal to  $7.4\% + (-0.9) \times 1\% + 1.3 \times 0.5\% = 7.15\%$ .

***FRM Part 1, Book 1: Foundations of Risk Management, Chapter 6: Arbitrage Pricing Theory and Multifactor Models of Risk and Return, Learning objective: Describe the inputs (including factor betas) to a multifactor model.***

3. Which of the following cases include a violation of GARP's rules of conduct?

- A. Frank Benton - a risk manager and participant in the FRM program - oversees all investments in the healthcare startups for an investment bank. After thorough analysis and evaluating all necessary disclosures, Benton approves a significant loan facility to LLL Medicine (a young startup initiated by Frank's brother).
- B. Andrea Jenkins, FRM, is a rising star at Trust Hedge Fund. Behind Jenkins' success is her ability to fully concentrate on investment analysis since all other concerns – like the evaluation of an investment's suitability based on funds' internal rules, verification of compliance with laws, etc. – are handled by a third-party accounting firm.
- C. Elisabeth Stone, FRM, is responsible for credit risk analysis of corporate borrowers in one of Mississauga's local banks. On a quarterly basis, she presents to the board of directors a summary of the key ratios used for identifying the credit quality of existing borrowers with large exposures. The analysis is based on quarterly reports published by the clients. In addition, she always includes one slide with her own opinion on changes in ratios presented in case of adverse market movements. The slide has a footnote clarifying that the presented information is based on Stone's estimations.
- D. Jack Merton, who recently passed part one of the FRM exam, was reviewing his analysis on interest rate risk presented to the ALCO committee of Warsaw Bank last week. He noticed a major mistake and immediately informed all parties involved in this mistake.

Correct answer: B

Explanation: GARP Members should comply with all applicable laws, rules, and regulations. A delegation of those responsibilities violates the GARP's rules of conduct.

Option A is incorrect. Benton prepared a detailed analysis and made all necessary disclosures, giving his employer the chance to decide on whether Benton's approval was biased or not.

Option C is incorrect. Members may present their own opinion as long as it is clearly separated from factual information.

Option D is incorrect. Members shall not knowingly misrepresent any information. In cases where the misrepresentation was not intentional, members are not in violation of rules.

***FRM Part 1, Book 1: Foundations of Risk Management, Chapter 10: GARP Code of Conduct, Learning objective: Describe the potential consequences of violating the GARP Code of Conduct***

4. Which of the following is (are) not (an) assumption(s) of the capital pricing model?

- I. Investors require higher returns as a pre-condition for higher risks
  - II. All investors have the same one-period time horizon
  - III. CAPM considers taxes but does not consider transaction costs
  - IV. Investors' only concern should be a systematic risk
- A. II only
  - B. III only
  - C. I and IV only
  - D. III and IV only

Correct answer: B

Explanation: Point III is incorrect. CAPM assumes there are no taxes and no transaction costs.

Points I and II are correct. CAPM assumes that the investors' time horizon is the same and that, having discounted future cash flows by the same discount rate (expected return), investors reach the same intrinsic price for each stock.

Point IV is correct. Beta risk is the systematic risk, and, under the perfect capital market, investors' only concern should be a systematic risk as it cannot be eliminated via diversification.

***FRM Part 1, Book 1: Foundations of Risk Management: Chapter 5- Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM), Learning objective: Describe the assumptions underlying the CAPM.***

5. Which of the following are the roles of the audit committee?

- I. Members must have the knowledge of basic accounting principles under US GAAP and IFRS
  - II. Ensuring that the remuneration of key management aligns with the goals of other stakeholders
  - III. Ensuring reasonable accuracy of the organization's financial statements
  - IV. Ensuring that management strategies are focused on economic profits
  - V. Ensuring that the organization has taken measures to avoid misstated errors and fraud
- A. I, II and IV
  - B. III, IV, and V
  - C. I, III and V
  - D. II, III and IV

Correct answer: C

Explanation: Statements II and IV are not the roles of the audit committee. It is the responsibility of the compensation committee to ensure that the remuneration of key management aligns with the goals of other stakeholders. The risk committee is responsible for ensuring that strategies put forth by the management are focused on economic profits rather than accounting profits. Statements I, III, and V are core responsibilities of the audit committee.

***FRM Part 1, Book 1: Foundations of Risk Management, Chapter 3: The Governance of Risk Management, Learning objective: Assess the role and responsibilities of a firm's audit committee.***

6. Which of the following statements inappropriately defines the elements of Enterprise Risk Management (ERM)?

I. Stakeholders management refers to sharing or communicating the information regarding a firm's internal risk management processes to external stakeholders.

II. Data technology and resources refer to the management of activities that are directly related to the production of products and services of the firm.

- A. I only
- B. II only
- C. Both I & II
- D. None of the above

Correct answer: B

Explanation: Point I is accurate. Stakeholder management refers to sharing or communicating the information regarding a firm's internal risk management process to external stakeholders, i.e., shareholders, creditors, regulators, and the public.

Statement II is inaccurate because the "Data technology and resources" element of ERM refers to the improvement of the quality and technology of data used in the evaluation of risks.

***FRM Part 1, Book 1: Foundations of Risk Management, Chapter 8: Enterprise Risk Management and Future Trends, Learning objective: Describe important dimensions of an ERM program and relate ERM to strategic planning.***

7. Enterprise-wide VaR is not likely to account for all types of risk. In particular, enterprise-wide VaR may not factor in:

- A. Credit risk
- B. Market risk
- C. Funding liquidity risk
- D. None of the above. Enterprise-wide risk is likely to account for all types of risk.



Correct answer: C

Explanation: Banks often categorize risks into three broad types: market risk, credit risk, and operational risk. However, enterprise-wide VaR is not likely to account for all three, especially operational risk. In addition, enterprise-wide VaR may not factor in funding liquidity risk - the risk that a sudden reduction in funding to the bank might force the sale of assets under duress, leading to losses.

**FRM Part 1, Book 1: Foundations of Risk Management, Chapter 3: The Governance of Risk Management, Learning objective: Evaluate the relationship between a firm's risk appetite and its business strategy, including the role of incentives.**

8. There are important similarities between the 2007-2009 crisis and the “The Big Five Crisis” of Spain, Norway, Finland, Sweden, and Japan. In all those crises, there were:

- I. Notable increases in public debt
  - II. Positive current account balances
  - III. Excessive borrowing from foreign institutions and governments
  - IV. House price run-ups
- A. I and IV
  - B. I, III and IV
  - C. II, III and IV
  - D. All of the above

Correct answer: B

Explanation: Post-2008-2009 crisis research and analysis have established that the financial crisis was “not special,” but was, in fact, reminiscent of prior crises. In particular, there are important similarities between the 2007-2009 crisis and the “The Big Five Crises” of Spain, Norway, Finland, Sweden, and Japan. In all those crises, there were (IV) house price run-ups, (I) notable increases in public debt, and (III) excessive borrowing from foreign institutions and governments, (II) triggering **negative** current account balances.

**FRM Part 1, Book 1: Foundations of Risk Management, Chapter 10: Anatomy of the Great Financial Crisis of 2007-2009, Learning objectives:**

- **Describe the historical background and provide an overview of the 2007-2009 financial crisis.**
- **Describe the build-up to the financial crisis and the factors that played an important role.**
- **Compare the roles of different types of institutions in the financial crisis, including banks, financial intermediaries, mortgage brokers and lenders, and rating agencies.**

9. Malala Pham is an equity analyst. She has been asked to derive the beta of a stock from the CAPM. What is the value of the beta if the risk-free rate is 3%, the expected return of the market is 14%, and the return on the stock is 11.8%?

- A. 0.63
- B. 1.5
- C. 0.8
- D. 1.07

Correct answer: C

Explanation: Expected return on the stock = Risk-free rate + Beta (Market risk - Risk-free rate)

$$11.8\% = 3\% + \beta \times (14\% - 3\%)$$

$$\beta = \frac{0.118 - 0.03}{0.14 - 0.03} = 0.8$$

**FRM Part 1, Book 1: Foundations of Risk Management, Chapter 5: Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM), Learning objective: Interpret beta and calculate the beta of a single asset or portfolio.**

10. A stock is trading at USD 35.12, and analysts expect its price in exactly 12 months to be USD 38.77. If the market risk is 8%, the risk-free rate is 5%, and the Beta of the stock is 1.8, the stock is most likely:

- A. Undervalued by USD 3.16
- B. Undervalued by USD 6.81
- C. Undervalued by USD 3.16
- D. Fairly valued

Correct answer: D

Explanation: Expected return on the stock = Risk-free rate + Beta (Market risk - Risk-free rate)

$$= 5\% + 1.8 \times (8\% - 5\%)$$

$$= 10.4\%$$

$$\text{Expected price} = \text{USD } 35.12 \times 1.104 = \text{USD } 38.77 = \text{Analyst estimates of USD } 38.77$$

**FRM Part 1, Book 1: Foundations of Risk Management, Chapter 5: Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM), Learning objective: Apply the CAPM in calculating the expected return on an asset**

11. The exhibit below summarizes risk and return for three market-neutral hedge funds in Canada:

**Exhibit**  
**Risk, Return and Fee Data for Three Market-Neutral Hedge Funds**

|                               | Fund A | Fund B | Fund C |
|-------------------------------|--------|--------|--------|
| Annualized return             | 15%    | 22%    | 9%     |
| Annualized standard deviation | 20%    | 36%    | 15%    |

If the risk-free rate in Canada is 2%, which of these funds is most appropriate from an investment perspective, according to the Sharpe ratio?

- A. Fund A
- B. Fund B
- C. Fund C
- D. Insufficient information

Correct answer: A

Explanation: To determine which fund is most appropriate for investment, the Sharpe ratio of the three funds is calculated as follows:

$$\text{Fund A} = \frac{15\% - 2\%}{20\%} = 0.65$$

$$\text{Fund B} = \frac{22\% - 2\%}{36\%} = 0.55$$

$$\text{Fund C} = \frac{9\% - 2\%}{15\%} = 0.47$$

Fund A has the highest Sharpe ratio, which means that it will enhance risk-adjusted-performance. As a result, it is the most suitable from an investment perspective.

**FRM Part 1, Book 1: Foundations of Risk Management, Chapter 5: Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM), Learning objective: Calculate, compare, and interpret the following performance measures: the Sharpe performance index, the Treynor performance index, the Jensen performance index, the tracking error, information ratio, and Sortino ratio.**

12. Which of the following risk metrics measures the risk as downside deviation instead of standard deviation?

- A. Treynor ratio
- B. Sharp ratio
- C. Sortino ratio
- D. Jensen measure

Correct answer: C

Explanation: The Sortino ratio uses the downside deviation as its risk measure. This addresses the problem of using standard deviation used by other risk measures since upside volatility is beneficial to investors.

**FRM Part 1, Book 1: Foundations of Risk Management, Chapter 5: Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM), Learning objective: Calculate, compare, and interpret the following performance measures: the Sharpe performance index, the Treynor performance index, the Jensen performance index, the tracking error, information ratio, and Sortino ratio.**

13. The common stock of General Electric is examined with a single factor model using unexpected percentage changes in GDP as the single factor. You have been provided with the following data:

- Expected return for General Electric = 7%
- GDP factor-beta = 0.8
- Expected GDP growth = 4%

Revised macroeconomic information strongly suggests that the GDP will grow by 2% as opposed to the original prediction of 4%. Assuming there's no new information regarding firm-specific events, the revised expected return using a single factor model is closest to:

- A. 5.9%
- B. 6.2%
- C. 8.6%
- D. 5.4%

Correct answer: D

Explanation: The equation for a single factor model for stock i is given by:

$$R_i = E(R_i) + \beta_{i,j} F_j + e_i$$

Where:

$R_i$  = revised return for stock i

$E(R_i)$  = the expected return for stock i

$\beta_{i,j}$  = the jth factor-beta for stock i

$F_j$  = the deviation of the factor j from its expected value

$e_i$  = the firm-specific return for stock i

In our case:

$$R_i = 0.07 + 0.8(0.02 - 0.04) = 0.054 \text{ or } 5.4\%$$

***FRM Part 1, Book 1: Foundations of Risk Management, Chapter 6: Multifactor Models of Risk-Adjusted Asset Returns, Learning objective: Calculate the expected return of an asset using a single-factor and a multifactor model.***

14. Consider a single factor APT in a country where the risk-free rate is 8%. The stocks of XENA have a beta of 1.7 and an expected return of 19%. The stocks of YNN have a beta of 2.2 and an expected return of 15%. Assuming you wanted to exploit an arbitrage opportunity, you would take a short position in:

- A. YNN and use the proceeds to take a long position in XENA
- B. XENA and use the proceeds to take a long position in YNN
- C. YNN and use the proceeds to take a long position in the riskless asset
- D. XENA and use the proceeds to take a long position in the riskless asset

Correct answer: A

Explanation: For XENA,  $19\% = 1.7F + 8\%$ ;  $F = 6.47\%$

For YNN,  $15\% = 2.2F + 8\%$ ;  $F = 3.18\%$

The single factor APT implies a higher expected return for XENA versus YNN, so XENA is underpriced and a buy opportunity. On the other hand, YNN is overpriced so shorting YNN and investing proceeds to XENA produces arbitrage profit as the factor (market premium assumed) is realized over time.

Thus, short YNN and take a long position in XENA.

Shorting stocks with positive expected values above the risk-free rate and taking a long position in the risk-free rate is not a viable option in this case.

***FRM Part 1, Book 1: Foundations of Risk Management, Chapter 6: Multifactor Models of Risk-Adjusted Asset Returns, Learning objective: Calculate the expected return of an asset using a single-factor and a multifactor model***

15. Jared Bosh, FRM, serves as a portfolio manager at Cali Bank. Among his clients is Gustavo Hernandez, a talented chef and restaurant owner, for whom Bosh manages a personal portfolio of stocks and government bonds.

Hernandez recently disclosed to Bosh that he is under investigation for being an illegal immigrant and might get deported in the coming months. Later that day, Bosh shares that information with his colleague at Cali Bank responsible for handling the business loans of Hernandez.

Jared Bosh has most likely:

- A. Not violated the GARP Code of Conduct because she revealed illegal activities on the part of her client
- B. Not violated the GARP Code of Conduct with respect to confidentiality because it is central to the bank's activities to know that a customer with outstanding loans might get deported
- C. Violated the GARP Code of Conduct by failing to detect the immigration status of his client
- D. Violated the GARP Code of Conduct with respect to confidentiality

Correct answer: D

Explanation: According to the GARP code of Conduct, members shall take “all reasonable precautionary measures to prevent intentional and unintentional disclosure of confidential information.”

Jared Bosh did not have the right to inform his colleagues at the bank of his client’s situation, even though Hernandez might have outstanding loans.

***FRM Part 1, Book 1: Foundations of Risk Management, Chapter 11: GARP Code of Conduct, Learning objectives:***

- ***Describe the responsibility of each GARP Member with respect to professional integrity, ethical conduct, conflicts of interest, confidentiality of information and adherence to generally accepted practices in risk management.***
- ***Describe the potential consequences of violating the GARP Code of Conduct.***

16. Which of the following is not included in the GARP Rules of Conduct?

GARP members shall:

- A. Act professionally, ethically and with integrity in all dealings with employers, existing or potential clients, the public, and other practitioners in the financial services industry.
- B. Not knowingly misrepresent details relating to analysis, recommendations, actions, or other professional activities.
- C. Not make use of confidential information for inappropriate purposes and unless having received prior consent shall maintain the confidentiality of their work, their employer or client.
- D. Act with honesty, integrity, and competence to fulfill the risk professional’s responsibilities and to uphold the reputation of the risk management profession.

Correct answer: D

Explanation: Options A, B, and C can be found in the GARP Rules of Conduct, but Option D is found in the Principles section under 1.1 Professional Integrity and Ethical Conduct.

***FRM Part 1, Book 1: Foundations of Risk Management, Chapter 14: GARP Code of Conduct, Learning objectives:***

- ***Describe the responsibility of each GARP Member with respect to professional integrity, ethical conduct, conflicts of interest, confidentiality of information and adherence to generally accepted practices in risk management.***
- ***Describe the potential consequences of violating the GARP Code of Conduct.***

17. Xiao Bing, head of compliance at OceanAsian Bank, quoted a key governance principle related to risk data aggregation provided by the Basel Committee, which states that “a bank should be able to generate aggregate risk data to meet a broad range of on-demand, ad hoc risk management reporting requests, including requests during stress/crisis situations, requests due to changing internal needs and requests to meet supervisory queries”.

Which of the following principles is Bing referring to?

- A. Accuracy and Integrity
- B. Completeness
- C. Adaptability
- D. Timeliness

Correct answer: C

Explanation: According to the Basel Committee’s 6th Principle “Adaptability,” a bank should be able to generate aggregate risk data to meet a broad range of on-demand, ad hoc risk management reporting requests, including requests during stress/crisis situations, requests due to changing internal needs and requests to meet supervisory queries.

A bank’s risk data aggregation capabilities should be flexible and adaptable to meet ad hoc data requests, as needed, and to assess emerging risks. Adaptability will enable banks to conduct better risk management, including forecasting information, as well as to support stress testing and scenario analyses.

***FRM Part 1, Book 1: Foundations of Risk Management, Chapter 7: Risk Data Aggregation and Reporting Principles, Learning objective:***

- ***Describe the role that supervisors play in the monitoring and implementation of the risk data aggregation and reporting practices.***
- ***Distinguish practical instances of Basel Committee’s Principles for sound management of operational risk.***

18. A 5-year research has been conducted on 25 different hedge funds in the United States. The researchers have categorized the funds into 4 distinct hedge fund strategies:

| Fund strategy          | Average annual return | Standard deviation | Beta |
|------------------------|-----------------------|--------------------|------|
| Global Macro           | 13.5%                 | 18%                | 1.15 |
| Long-Short Equity      | 21.3%                 | 31%                | 3.1  |
| Equity arbitrage       | 22.6%                 | 35%                | 3.4  |
| Fixed income arbitrage | 8%                    | 2%                 | 0.6  |

If the risk-free rate is 4% and the S&P 500 has returned an average of 10.1% over the 5 year period, then which fund strategy has performed the best according to the Treynor measure?

- A. Global Macro
- B. Long-Short Equity
- C. Equity arbitrage
- D. Fixed income arbitrage

Correct answer: A

Explanation: The Treynor measure,  $T_p = \{E(R_p) - R_f\} / \beta_p$

Hence:

$$\text{Global Macro} = \frac{13.5 - 4}{1.15} = 8.26$$

$$\text{Long - Short Equity} = \frac{21.3 - 4}{3.1} = 5.58$$

$$\text{Equity arbitrage} = \frac{22.6 - 4}{3.4} = 5.47$$

$$\text{Fixed income arbitrage} = \frac{8 - 4}{0.6} = 6.67$$

**FRM Part 1, Book 1: Foundations of Risk Management, Chapter 5: Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM), Learning objective: Calculate, compare and interpret the following performance measures: the Sharpe performance index, the Treynor performance index, the Jensen performance index, the tracking error, information ratio and Sortino ratio.**

19. Which of the following best defines a nonrecourse loan?

- A. A type of loan secured by collateral in which the issuer can seize the collateral but cannot seek from the borrower for any further compensation if the borrower defaults
- B. A type of unsecured loan in which the issuer cannot seek out the borrower for any compensation if the borrower defaults
- C. A type of loan secured by collateral, usually a property, in which the issuer can only seek further compensation in the event the collateral does not cover the full value of the defaulted amount
- D. A type of unsecured loan in which the borrower pays a very low initial interest **rate** that increases after a few years

Correct answer: A

Explanation: Nonrecourse debt is a type of loan secured by collateral, which is usually a property. If the borrower defaults, the issuer can seize the collateral but cannot seek further compensation from the borrower, even if the collateral does not cover the full value of the defaulted amount.

In the lead-up to the financial crisis of 2007-2008, many nonrecourse mortgages allowed the lender only to take possession of the borrower's home. When the housing prices fell, the borrowers sold their homes



to the lender for the principal outstanding on the mortgage because the collateral did not cover the full value of the defaulted amount.

***FRM Part 1, Book 1: Foundations of Risk Management, Chapter 10. Anatomy of the Great Financial Crisis of 2007-2009, Learning objectives:***

- *Explain the role of subprime mortgages and collateralized debt obligations (CDOs) in the crisis.*
- *Compare the roles of different types of institutions in the financial crisis, including banks, financial intermediaries, mortgage brokers and lenders, and rating agencies.*

20. According to the Capital Asset Pricing Model (CAPM), which of the following risks should an investor be compensated for?

I. Unsystematic risk

II. Asset-specific risk

III. Systematic risk

- A. I only
- B. III only
- C. I and II only
- D. I and III only

Correct answer: B

Explanation: The investor should only be compensated for systematic risk or beta. Since CAPM assumes that there are no transaction costs, and an investor can diversify free of cost, the investor does not need to be compensated for diversifiable risks, i.e., unsystematic or asset-specific risk.

***FRM Part 1, Book 1: Foundations of Risk Management, Chapter 5. Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM), Learning objectives:***

- *Describe the assumptions underlying the CAPM.*
- *Interpret beta and calculate the beta of a single asset or portfolio.*

### Quantitative Analysis

21. An analyst is performing a regression analysis on the daily prices of Google (GOOG) compared to the prices of the S&P 500 index. Below is the data that the analyst managed to collect for the last 5 trading days:

| Closing Price | GOOG  | S&P   |
|---------------|-------|-------|
| Dec 05, 2017  | 1,005 | 2,630 |
| Dec 04, 2017  | 999   | 2,639 |
| Dec 01, 2017  | 1,010 | 2,642 |
| Nov 30, 2017  | 1,021 | 2,648 |
| Nov 29, 2017  | 1,022 | 2,626 |

What is the equation for the regression line?

- A.  $y = 995 + 0.00625x$
- B.  $y = 574 + 0.859x$
- C.  $y = 0.859 + 574x$
- D.  $y = 989 + 0.859x$

Correct answer: A

Explanation: First, we need to estimate regression coefficients.

$$b_1 = \frac{\text{Cov}(X,Y)}{\text{Var}(X)} = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2} = \frac{2}{320} = 0.00625$$

$$b_0 = \bar{Y} - b_1 \times \bar{X} = 1,011.40 - 0.00625 * 2,637 = 995$$

The equation of regression line is

$$y = 995 + 0.00625x$$

| Close Price    | Y (GOOG)        | X (S&P)       | $Y_i - \bar{Y}$ | $X_i - \bar{X}$ | $(X_i - \bar{X}) \times (Y_i - \bar{Y})$ | $(X_i - \bar{X})^2$ |
|----------------|-----------------|---------------|-----------------|-----------------|------------------------------------------|---------------------|
| Dec 05, 2017   | 1,005           | 2,630         | -6.4            | -7              | 44.8                                     | 49                  |
| Dec 04, 2017   | 999             | 2,639         | -12.4           | 2               | -24.8                                    | 4                   |
| Dec 01, 2017   | 1,010           | 2,642         | -1.4            | 5               | -7                                       | 25                  |
| Nov 30, 2017   | 1,021           | 2,648         | 9.6             | 11              | 105.6                                    | 121                 |
| Nov 29, 2017   | 1,022           | 2,626         | 10.6            | -11             | -116.6                                   | 121                 |
| <b>Sum</b>     | <b>5,057</b>    | <b>13,185</b> | <b>0</b>        | <b>0</b>        | <b>2</b>                                 | <b>320</b>          |
| <b>Average</b> | <b>1,011.40</b> | <b>2,637</b>  |                 |                 |                                          |                     |

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 7: Linear Regression, Learning objectives:**

- **Interpret the results of an ordinary least squares (OLS) regression with a single explanatory variable.**
- **Describe the key assumptions of OLS parameter estimation.**

22. Luke Friday, FRM, runs a consultancy firm that offers investment advice to clients around Los Angeles. The number of clients the firm receives in a month is distributed as a Poisson variable with a mean of 2. What is the probability that the firm receives exactly 30 new clients in a year, assuming every client is independent?

- A. 0.025
- B. 0.0363
- C. 0.24
- D. 0.00363

Correct answer: B

Explanation: We note the following:

- I. New clients are received randomly at a rate of 2 per unit time (month)
- II. Each event is independent

These observations confirm the Poisson distribution.

Since the question asks us to find a yearly probability, we must convert the monthly Poisson rate to an equivalent yearly rate.

Poi(2) monthly distribution is equivalent to Poi(24)

Now, if  $X$  stands for the number of clients received,

$$P(X = x) = [\exp(-\lambda) \lambda^x] / x!$$

$$P(X = 30) = [\exp(-24) 24^{30}] / 30!$$

$$= 0.0363$$

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 3: Common Univariate Random Variables, Learning objective: Distinguish the key properties and identify the common occurrences of the following distributions: uniform distribution, Bernoulli distribution, binomial distribution, Poisson distribution, normal distribution, lognormal distribution, Chi-squared distribution, Student's  $t$ , and  $F$ -distributions.**

23. A portfolio manager has recently invested in two stocks – HKJ and YYG. The probability that HKJ posts a return greater than 10% this year is 0.35, and the probability that YYG posts a return greater than 10% is 0.45.

Given that the probability that both stocks post returns greater than 10% this year is 0.15, what is the probability that either HKJ or the YYG or both stocks post returns greater than 10% this year?

- A. 0.75
- B. 0.55
- C. 0.95
- D. 0.65

Correct answer: D

Explanation: Let  $A = \{\text{HKJ overperforms 10\%}\}$  and  $B = \{\text{YYG overperforms 10\%}\}$

Therefore,  $P(A) = 0.35$ ,  $P(B) = 0.45$ , and  $P(A \cap B) = 0.15$

We want to determine  $P(A \cup B)$ ;

The addition rule states that  $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Hence,  $P(A \cup B) = 0.35 + 0.45 - 0.15 = 0.65$

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 1: Fundamentals of Probability, Learning objectives:**

- **Calculate the probability of an event for a discrete probability function.**
- **Define and calculate a conditional probability**

24. A retail investor has recently bought two stocks – UUK and LLY – which have a correlation of 0.72. The covariance between UUK and LLY is 0.032.

Given that LLY's return has a standard deviation of 0.17, the variance of returns for UUK is closest to:

- A. 0.13
- B. 0.06835
- C. 0.0385
- D. 0.0148

Correct answer: B

Explanation: Correlation between UUK and LLY,

$$\text{Corr}(UUK, LLY) = \text{Cov}(UUK, LLY) / (\sigma_{UUK} \times \sigma_{LLY})$$

$$0.720 = \frac{0.032}{\sigma_X \times 0.17}$$

$$0.1224\sigma_X = 0.032$$

$$\sigma_X = 0.26144$$

$$V(X) = 0.26144^2 = 0.06835$$

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 2: Multivariate Random Variables, Learning objectives:**

***Explain the relationship between the covariance and correlation of two random variables and how these are related to the independence of the two variables.***

***Explain the effects of applying linear transformations on the covariance and correlation between two random variables. Compute the variance of a weighted sum of two random variables.***

25. The FRM part 1 exam attracts candidates from all over the world. A quantitative analyst compiled the following statistics:

| Country of residency | Probability of passing the FRM Part 1 exam | A portion of total candidates |
|----------------------|--------------------------------------------|-------------------------------|
| United States        | 0.40                                       | 0.22                          |
| India                | 0.42                                       | 0.21                          |
| Canada               | 0.39                                       | 0.14                          |
| Other                | 0.31                                       | 0.44                          |

If a randomly selected candidate has just passed the FRM part 1 exam, the probability that the candidate comes from India is closest to:

- A. 22%
- B. 24%
- C. 25%
- D. 9%

Correct answer: B

Explanation: We wish to determine  $P(C_2 | C)$

$$\begin{aligned} P(C_2 | C) &= \frac{(P(C_2) \times P(C | C_2))}{[P(C_2) \times P(C | C_2) + (P(C_1) \times P(C | C_1)) + (P(C_3) \times P(C | C_3)) + (P(C_4) \times P(C | C_4))]} \\ &= \frac{0.21 \times 0.42}{[(0.21 \times 0.42) + (0.22 \times 0.40) + (0.14 \times 0.39) + (0.44 \times 0.31)]} \\ &= \frac{0.0882}{0.0882 + 0.088 + 0.0546 + 0.1364} \\ &= 24.02\% \end{aligned}$$

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 1: Fundamentals of Probability, Learning objective: Define and calculate a conditional probability.**

26. A random sample of 120 portfolio managers in the United States was found to have an average salary of USD 72,000. The standard deviation among portfolio managers is known as approximately USD 7,800.

Assuming that salaries follow a normal distribution, what is the 2-sided 95% confidence interval for the mean salary of FRM portfolio managers in the United States?

- A. (USD 70,162.94, USD 73,837.06)
- B. (USD 56,712, USD 87,288)
- C. (USD 71,872.60, USD 72,127.40)
- D. (USD 70,604.40, USD 73,395.60)

Correct answer: D

Explanation: For any sample that comes from a normally distributed population, we know that:

$(\bar{x} - \mu)/(\sigma/\sqrt{n}) \sim N(0,1)$  when  $\sigma$  is known

Thus, a 95% CI for the mean,  $\mu = \bar{x} \pm Z_{\alpha/2} \times \sigma/\sqrt{n}$  where  $\alpha = 0.05$

$$= \text{USD } 72,000 \pm 1.96 \times \text{USD } 7,800/\sqrt{120}$$

$$= \text{USD } 72,000 \pm \text{USD } 1,395.60$$

Which can be expressed as (USD 70,604.40, USD 73,395.60).

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 6: Hypothesis Testing, Learning objectives:**

- Differentiate between a one-sided and a two-sided test and identify when to use each test.
- Construct and apply confidence intervals for one-sided and two-sided hypothesis tests and interpret the results

27. A podiatrist has the following data regarding the height of pre-teenagers:

|                       |     |     |     |
|-----------------------|-----|-----|-----|
| Age in years          | 11  | 12  | 13  |
| Height in centimeters | 144 | 150 | 157 |

Further information:  $S_{XX} = 66$ ,  $S_{YY} = 3.11$ ,  $S_{XY} = 11.3$

The Y-intercept is closest to:

- A. 148.27
- B. 106.73
- C. -104.33
- D. 80.24

Correct answer: A

Explanation: Under OLS estimation,

The regression equation is of the form:  $y_i = \alpha + \beta x_i$

Where Y is the dependent variable (height), X is the independent variable (age),  $\alpha$  = the y-intercept, and  $\beta$  = the slope.

$$\beta = \frac{\sum \frac{(x_i - \bar{x})(y_i - \bar{y})}{(x_i - \bar{x})^2}}{S_{xx}} = \frac{S_{xy}}{S_{xx}} = \frac{11.3}{66} = 0.1712$$

$$\alpha = \bar{y} - \beta \bar{x}$$

$$\bar{y} = \frac{144 + 150 + 157}{3} = 150.33$$

$$\bar{x} = \frac{11 + 12 + 13}{3} = 12$$

$$\text{Therefore, } \alpha = 150.33 - 0.1712 \times 12 = 148.27$$

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 7: Linear Regression, Learning objectives:**

- **Interpret the results of an ordinary least squares (OLS) regression with a single explanatory variable.**
- **Describe the key assumptions of OLS parameter estimation.**

28. The estimated slope coefficient ( $\beta_1$ ) for a certain stock is 0.6566, with a standard error equal to 0.0799. The sample had 10 observations, and a researcher wants to know if the slope coefficient is statistically different than zero. Using a 5% level of significance, what are the test statistic and the decision rule?

- A. 0.1217; do not reject  $H_0$
- B. 8.2177; do not reject  $H_0$
- C. 0.1217; reject  $H_0$
- D. 8.2177; reject  $H_0$

Correct answer: D

Explanation: The first step entails formulating the hypothesis:

$H_0: \beta_1 = 0$  vs.  $H_a: \beta_1 \neq 0$

The test statistic for the slope coefficient takes the form:  $t_{(\alpha, n-2)} = \frac{\beta_1 - \beta_0}{se(\beta_1)}$

In this case,  $t_{0.025, 8} = \frac{0.6566 - 0}{0.0799} = 8.2177$

( $\alpha = \frac{5\%}{2}$  since this is a 2-tailed test)

The critical 2-tailed values (  $t_{0.025, 8}$  ) are  $\pm 2.306$

Our test statistic lies outside the non-rejection region (-2.306, 2.306). As such, we have sufficient evidence to reject the null hypothesis and conclude that the slope coefficient is statistically different than zero.

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 7: Linear Regression, Learning objectives:**

- **Construct, apply and interpret hypothesis tests and confidence intervals for a single regression coefficient in a regression.**
- **Explain the steps needed to perform a hypothesis test in linear regression.**

29. Which of the following are assumptions of the multiple linear regression model?

- I. The error term has an expected value equal to zero and is normally distributed
  - II. The error for one observation is uncorrelated with that of a different observation
  - III. A non-linear relationship exists between the dependent and independent variables
  - IV. The independent variables are random, and there is an exact linear relationship between any two or more independent variables
- A. I and II
  - B. I and III



- C. II and IV
- D. III and IV

Correct answer: A

Explanation: The following are assumptions of the multiple linear regression model:

- (I) The error term has an expected value equal to zero and is normally distributed
- (II) The error for one observation is uncorrelated with that of a different observation
- (III) A linear relationship exists between the dependent and independent variables
- (IV) The independent variables must not be random, and there should be no exact linear relationship between any two or more independent variables.

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 8: Regression with Multiple Explanatory Variables, Learning objective: Distinguish between the relative assumptions of single and multiple regressions.**

30. An analyst believes that future 15-year real earnings of the S&P 500 are a function of the trailing dividend payout ratio of the stocks in the index (*DB*) and the yield curve slope (*YC*). She collects data and obtains the following multiple regression results:

|                  | Coefficient | Standard error |
|------------------|-------------|----------------|
| <b>Intercept</b> | -15.2%      | 3.589%         |
| <b>DB</b>        | 0.37        | 0.099          |
| <b>YC</b>        | 0.18        | 0.133          |

If the number of observations is assumed to be 1003, test the statistical significance of the independent variable *DB* at the 1% level of significance, quoting the value of the test statistic and the conclusion.

(If needed, refer to the t-table at the end of the document.)

- A. Test statistic = 0.2676; The DB regression coefficient is statistically different from zero
- B. Test statistic = 3.7373; The DB regression coefficient is statistically different from zero
- C. Test statistic = 0.2676; The DB regression coefficient is not statistically different from zero
- D. Test statistic = 3.7373; The DB regression coefficient is not statistically different from zero

Correct answer: B

Explanation: We are testing the following hypothesis:

$$H_0: DB = 0 \text{ vs } H_1: DB_1 \neq 0$$

The test statistic is  $0.37/0.099 = 3.7373$

$$t_{\alpha/2, n-k-1} = t_{0.01, 1000} = 2.330$$

The test statistic (3.7373) is greater than the upper critical value (2.330) of the t-distribution with 1000 degrees of freedom. Therefore, we reject the null hypothesis and conclude that the DB regression coefficient is statistically different from zero at the 1% level of significance.

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 8: Regression with Multiple Explanatory Variables, Learning objective: Construct, apply and interpret joint hypothesis tests and confidence intervals for multiple coefficients in a regression**

31. Suppose we have the following random variables:

{12 13 54 56 25}

Determine the skewness and kurtosis of the sample data.

- A. Kurtosis=0.7861, Skewness=0.1835
- B. Kurtosis=0.1861, Skewness=0.1865
- C. Kurtosis=0.7661, Skewness=0.1885
- D. Kurtosis=0.8861, Skewness=0.1855

The correct answer is: A

Explanation: We know that:

$$Kurt(X) = \frac{\mu_4}{\sigma^4}$$

Where

$$\mu_4 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^4$$

Which can be estimated as:

$$\widehat{Kurt}X = \frac{\hat{\mu}_4}{\hat{\sigma}^4} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^4}{\hat{\sigma}^4}$$

Analogously, the Skewness is given by:

$$\widehat{Skew}(X) = \frac{\hat{\mu}^3}{\hat{\sigma}^3} = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^3}{\hat{\sigma}^3}$$

First, we must determine the sample mean and the sample standard deviation of the data:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{1}{5} (12 + 13 + 54 + 56 + 25) = \frac{160}{5} = 32$$

For the variance, we have to calculate the unbiased estimator:

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2 = \frac{(12-32)^2 + \dots + (25-32)^2}{5-1} = 467.5$$

So that the standard deviation is given by:

$$\sqrt{467.5} = 21.62467.5 = 21.62$$

So,

$$Kurt(X) = \frac{\frac{1}{5}((12-32)^4 + \dots + (25-32)^4)}{21.62^4} = \frac{105395.60}{21.62^4} = 0.7861$$

And

$$skew(X) = \frac{\frac{1}{5}[(12-32)^3 + \dots + (25-32)^3]}{21.62^3} = 0.1835$$

Since the Skewness is positive, then the data is positively skewed (which is evident in the dataset given).

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 5: Sample moments, Learning objective: Estimate and interpret the skewness and kurtosis of a random variable**

32. Caterpillar sales were USD 45.5 billion for the year 2017. The following table presents the sales for each of the four quarters of 2017:

| Quarter                 | Q1   | Q2   | Q3  | Q4  |
|-------------------------|------|------|-----|-----|
| Sales (in billion USD ) | 15.2 | 17.2 | 8.3 | 4.8 |

Which of the following is the seasonal index for Q2 of 2017?

- A. 1.38
- B. 0.38
- C. 0.51
- D. 1.51

Correct answer: D

Explanation: The norm (average monthly sales) = USD 45.5 billion/4 quarters = USD 11.375 billion/quarter

The seasonal index = USD 17.2 billion/USD 11.375 billion = 1.51

An index of 1.51 means that January sales are 51% above the norm.

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 11: Non-stationary Time Series, Learning objectives:**

- Explain how to construct an *h*-step-ahead point forecast for a time series with seasonality.
- Calculate the estimated trend value and form an interval forecast for a time series

33. The following sample autocorrelation estimates are obtained using 333 data points:

|                    |      |       |       |
|--------------------|------|-------|-------|
| <b>Lag</b>         | 1    | 2     | 3     |
| <b>Coefficient</b> | 0.12 | -0.17 | -0.09 |

The value of the Ljung Box Q-statistic is closest to:

- A. 17.32
- B. -47.40
- C. 0.05
- D. 17.12

Correct answer: A

Explanation: The Ljung Box statistic, also known as the modified Box Pierce statistic, is a function of the accumulated autocorrelations,  $\rho_j$ , up to time lag  $m$ . It's calculated as:

$$Q(m) = n(n + 2) \sum \left[ \frac{\rho_j^2}{n - j} \right]$$

In this case, time lag = 3

$$\begin{aligned} \text{Thus, } Q(3) &= 333(335) \left[ \frac{0.12^2}{332} + \frac{(-0.17)^2}{331} + \frac{(-0.09)^2}{330} \right] = 17.32 \\ &= -0.3619 \text{ or } -2.7630 \end{aligned}$$

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 10: Stationary Time Series, Learning objectives:**

- **Define the autocovariance function and the autocorrelation function**
- **Describe the Box-Pierce Q-statistic and the Ljung-Box Q statistic**
- **Explain how a lag operator works.**

34. The estimated standard error for a Monte Carlo simulation without antithetic variables is 3.65. The antithetic variables are now included so that the correlation between the pairs is -0.65 and the simulation is repeated 100 times. What is the percentage change in standard error?

- A. 45.65%
- B. -46.65%
- C. 40.84%
- D. -40.84%

Correct answer: D

Explanation: Recall that the standard error of the simulation when antithetic variables are included expectation is given by:

$$\frac{\sigma_g \sqrt{1+p}}{\sqrt{b}}$$

We are told that,

$$\frac{\sigma_g}{b} = 3.65 \Rightarrow \sigma_q = 3.65 \times \sqrt{100} = 36.5$$

So the new standard error is given by:

$$\frac{\sigma_g \sqrt{1+p}}{\sqrt{b}} = \frac{36.5 \times \sqrt{1-0.65}}{\sqrt{100}} = 2.1593$$

So the percentage change is:

$$\frac{2.1593}{3.65} - 1 = -40.84\%$$

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 13: Simulation and Bootstrapping, Learning objective: Explain the use of antithetic and control variates in reducing Monte Carlo sampling error.**

35. A model is estimated as:

$$Y_i = 3 + 1.5X_{1i} - 2X_{2i} + \epsilon_i$$

Where  $\sigma_{X_1}^2 = 25$  and  $\sigma_{X_2}^2 = 36$ .

What is the value of  $\rho_{X_1X_2}$  given that  $\widehat{\beta}_1 = 1.7$  such that the model is reduced to:

$$Y_i = \alpha + \hat{\beta}X_{1i} + \epsilon_i$$

- A. 0.765
- B. -0.0754
- C. -0.635
- D. -0.0833

Correct answer: D

Explanation: Using the omitted variable formula, we know that:

$$\hat{\beta}_1 = \beta_1 + \beta_2\delta$$

Where:

$$\delta = \frac{Cov(X_1, X_2)}{Var(X_1)}$$

$$\hat{\beta}_1 \Rightarrow \beta_1 + \beta_2 \delta = 1.7$$

$$= 1.5 - 2 \left( \frac{\rho_{X_1 X_2} \sigma_{X_1 X_2}^2}{\text{Var}(X_1)} \right) = 1.7$$

$$\Rightarrow \rho_{X_1 X_2} = -0.0833$$

Note that we have used the analogy:

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \Rightarrow \text{Cov}(X, Y) = \rho_{XY} \sigma_X \sigma_Y$$

Where:

$\rho_{XY}$  = the correlation between X and Y

$\sigma_X$  = standard deviation X

$\sigma_Y$  = standard deviation of Y

$\text{Cov}(X, Y)$  = covariance between X and Y

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 9: Regression Diagnostic, Learning objective: Describe the consequences of excluding a relevant explanatory variable from a model and contrast those with the consequences of including an irrelevant regressor.**

36. The returns from a portfolio are as shown in the table below.

| Time | Price  |
|------|--------|
| 0    | 100    |
| 1    | 98.56  |
| 2    | 98.65  |
| 3    | 97.76  |
| 4    | 96.50  |
| 5    | 100.56 |

What is the value of continuously compounded return at time 3?

- A. 0.81%
- B. -0.81%
- C. 0.91%
- D. -0.91%

Correct answer: D

Explanation: The continuously compounded return is given by:

$$r_t = \ln P_t - \ln P_{t-1}$$

So that:

$$\begin{aligned} r_3 &= \ln p_3 - \ln P_2 \\ &= \ln 97.76 - \ln 98.65 = -0.91\% \end{aligned}$$

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 12: Measuring Return, Volatility, and Correlation, Learning objective: Calculate, distinguish, and convert between simple and continuously compounded returns.**

37. A CNBC analyst is estimating the future value of cryptocurrency wallets, where the mean ending value in USD is USD 95,000, the standard deviation is USD 19,300, and the number of simulations is 1000. Which of the following is closest to the 95% confidence interval he would find with this simulation?

- A. (USD 93,802.55, USD 96,197.45)
- B. (91,213.34, USD 98,786.66)
- C. (USD 94,962.13, USD 95,037.87)
- D. (USD 94,727.43, USD 95,272.57)

Correct answer: A

Explanation: The confidence interval is constructed using the student's t-distribution and the interval is given by:

$$X - 1.962 \times (s/\sqrt{N}), \bar{X} + 1.962 \times (s/\sqrt{N})$$

Thus,

$$\begin{aligned} \text{CI} &= \text{USD } 95,000 - 1.962(\text{USD } 19,300/\sqrt{1000}), \text{USD } 95,000 + 1.962(\text{USD } 19,300/\sqrt{1000}) \\ &= (\text{USD } 93,802.55, \text{USD } 96,197.45) \end{aligned}$$

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 6: Hypothesis Testing, Learning objective: Construct and apply confidence intervals for one-sided and two-sided hypothesis tests and interpret the results of hypothesis tests with a specific level of confidence.**

38. Which of the following statements is/are accurate?

The autoregressive moving average (ARMA) process forms an important part of time series analysis since it:

- I. Captures a very robust picture of the variable being estimated thanks to the exclusion of lagged random shocks and lagged observations
  - II. Combines the lagged unobservable random shock characteristic of the MA process with the observed lagged time series characteristic of the AR process
  - III. Involves gradually-decaying autocorrelations
- A. I & II only
  - B. II & III only
  - C. I & III only
  - D. All of the above

Correct answer: B

Explanation: Statement I is incorrect. The ARMA process forms an important part of time series analysis since it captures a very robust picture of the variable being estimated thanks to the inclusion of lagged random shocks and lagged observations.

Statement II is accurate. The ARMA process combines the lagged unobservable random shock characteristic of the MA process with the observed lagged time series characteristic of the AR process.

Statement III is accurate. The ARMA process involves gradually-decaying autocorrelations.

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 10: Stationary Time Series, Learning objectives:**

- **Define and describe the properties of autoregressive (AR) processes.**
- **Define and describe the properties of moving average (MA) processes.**
- **Explain how a lag operator works.**
- **Define and describe the properties of autoregressive moving average (ARMA) processes**

39. The CEO of AlphaTrynn Hedge Fund estimates that the effect of increasing the number of qualified financial analysts hired by one will improve the fund's annual return by 0.3% with a standard error of 0.12%. Assuming the fund's returns are normally distributed, which of the following best represents a 90% 2-sided confidence interval for the size of the slope coefficient?

- A. (0.12%, 0.45%)
- B. (0.27%, 0.61%)
- C. (0.20%, 0.54%)
- D. (0.10%, 0.50%)



Correct answer: D

Explanation: The CI will take the form:

$$\beta_1 \pm Z_{\alpha/2} \times \text{se}(\beta_1)$$

$$= \beta_1 \pm Z_{0.05} = 0.3\% \pm 1.645 \times 0.12\% = (0.10\%, 0.50\%)$$

**FRM Part 1, Book 2: Quantitative Analysis, Chapter 7: Linear Regression, Learning objectives:**

- **Construct, apply, and interpret hypothesis tests and confidence intervals for a single regression coefficient in a regression.**
- **Explain the steps needed to perform a hypothesis test in linear regression.**

40. The world health organization tracked a group of 50,000 Indians for 20 years. At the commencement of the study, 75% were categorized as omnivores, 15% as vegetarians, and 10% as vegans. Results revealed that vegetarians were twice as likely as vegans to die from colon cancer, but only half as likely as omnivores. During the period, a randomly selected group member passed on from colon cancer.

What is the probability that the individual who died was an omnivore?

- A. 88.23%
- B. 75%
- C. 83.33%
- D. 85.56%

Correct answer: A

Explanation: Let:

D = Event of death

VT = Event of vegetarians

OM = Event of omnivores

VG = Event of vegan

We need to calculate  $P(OM | D)$

Now, we already know that:

$$P(D | VT) = 2P(D | VG) \text{ and } P(D | VT) = 1/2P(D | OM)$$

Applying Bayes' theorem,

$$\begin{aligned} P(OM | D) &= \frac{P(OM) \times P(D | OM)}{(P(OM) \times P(D | OM) + P(VT) \times P(D | VT) + P(VG) \times P(D | VG))} \\ &= \frac{2P(D | VT) \times 0.75}{(2P(D | VT) \times 0.75) + P(D | VT) \times 0.15 + 1/2P(D | VT) \times 0.10} \end{aligned}$$

$$= \frac{1.5}{1.5 + 0.15 + 0.05}$$
$$= 0.8823$$

***FRM Part 1, Book 2: Quantitative Analysis, Chapter 1: Fundamentals of Probability, Learning objectives:***

- *Define and calculate a conditional probability.*
- *Distinguish between conditional and unconditional probabilities.*
- *Explain and apply Bayes' rule.*

## **Financial Markets and Products**

41. An exotic option where the payoff of the option is not based on the price of the underlying at the end of the contract, but on the arithmetic average of the prices during the life of the option is most likely called a/an:

- A. Gap option
- B. Lookback option
- C. Asian option
- D. Compound option

Correct answer: C

Explanation: Asian options are options where the payoff depends on the arithmetic average of the price of the underlying asset during the life of the option.

Option A is incorrect. A gap option is a type of binary option whose stated strike price is different from its payoff strike.

Option B is incorrect. A lookback option is an exotic option that allows investors to “look back” at the underlying prices occurring over the life of the option and then exercise based on the underlying asset’s optimal value.

Option D is incorrect. A compound option or split-fee option is an option on an option. The exercise payoff of a compound option involves the value of another option. A compound option then has two expiration dates and two strike prices.

***FRM Part 1, Book 3: Financial Markets and Products, Chapter 15: Exotic Options, Learning objective: Identify and describe the characteristics and payoff structure of the following exotic options: gap, forward start, compound, chooser, barrier, binary, lookback, Asian, exchange and basket options***

42. Karsley Bank, located in Arizona, wishes to establish its footing in Delaware by acquiring Quota Bank. Employees and senior management of the latter do not want their business to be enjoined to that of the former in part because they feel their bank’s future is bright. In particular, the directors of Quota Bank fear that the new owners may opt to fire them in favor of new management located in Arizona. The directors decide to seek advice from a reputable investment bank on how to fend off the takeover.

Assuming you were one of the advisory panel members, which of the following would likely **not** form part of your advice?

- A. That Quota Bank adds to its charter a provision that if another company acquires one-third of the shares, other shareholders have the right to sell their shares to that company for a large premium over market prices
- B. That Quota Bank should file a lawsuit to dispute the possible takeover
- C. That Quota Bank adds to its charter a provision making it impossible for any new owners to terminate the contracts of existing directors
- D. That Quota Bank grants its employees stock options that can be exercised in the event of a takeover

Correct answer: B

Explanation: Options A, C, and D are all possible and realistic actions that Quota Bank can take to discourage other companies from acquiring the bank.

However, going to court by filing a lawsuit might have little or no impact in light of the information given in the question. Takeover bids are legal procedures between a willing buyer and a willing seller. Legal proceedings would work only if the potential owners engage in outlawed activities in an attempt to pull off the acquisition.

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 1: Banks, Learning objectives:**

- **Describe investment banking financing arrangements, including private placement, public offering, best efforts, firm commitment, and Dutch auction approaches.**
- **Describe the potential conflicts of interest among commercial banking, securities services, and investment banking divisions of a bank and recommend solutions to the conflict of interest problems.**

43. The following data gives the mortality expectancy among women in India in 1954.

| Age in years | Probability of death within one year | Life expectancy |
|--------------|--------------------------------------|-----------------|
| 30           | 0.001312                             | 47.52           |
| 31           | 0.001345                             | 46.59           |
| 32           | 0.001383                             | 45.65           |
| 33           | 0.001415                             | 44.73           |

Which of the following is the probability of a woman who is aged 30 dying between ages 31 and 32?

- A. 0.001312
- B. 0.001345
- C. 0.001343
- D. 0.001348

Correct answer: C

Explanation: The probability of a woman who is aged 30 dying between ages 31 and 32 means the woman survives for one year but dies in the second year. This is equal to the probability that she does not die in the first year (between ages 30 and 31) multiplied by the probability that she does die in the second year (between ages 31 and 32). This is:

$$(1 - 0.001312) \times 0.001345 = 0.001343$$

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 2: Insurance Companies and Pension Plans, Learning objective: Describe the use of mortality tables and calculate the premium payment for a policyholder.**

44. Consider the following statements regarding closed-end mutual funds:

- I. The funds offer investors professional management
- II. Shares at times trade at a discount to the NAV
- III. Shares at times trade at a premium to the NAV
- IV. Funds are redeemed at their NAVs

Which of the statements above is/are incorrect?

- A. IV only
- B. II and III only
- C. II, III and IV
- D. All of the above

Correct answer: A

Explanation: Shares of closed-end mutual funds are redeemed at their prevailing market values, not at their NAVs.

All other statements are correct. Shares at times trade at a discount or at a premium from the NAV, and these funds offer investors professional management.

***FRM Part 1, Book 3: Financial Markets and Products, Chapter 3: Fund Management, Learning objective: Differentiate among open-end mutual funds, closed-end mutual funds, and exchange-traded funds (ETFs).***

45. Blackoil Inc. is an American oil producing company that regularly sells oil futures to reduce the risk of fluctuating oil prices. This activity can be described as:

- A. Speculating
- B. Hedging
- C. Clearing
- D. None of the above

Correct answer: B

Explanation: Futures contracts are one of the most common derivatives used to hedge risk. Hedgers enter into futures contracts to reduce the price risk of underlying assets. Hedging is an attempt to “lock in” the value of an asset and hence guarantee a certain level of return to the investor.

***FRM Part 1, Book 3: Chapter 4: Introduction to Derivatives, Learning objective: Differentiate among the broad categories of traders: hedgers, speculators, and arbitrageurs.***

46. Fatima Dow, a derivatives trader at HEC Hedge Fund, entered into a March silver futures contract on the New York Mercantile Exchange (NYMEX) to purchase 5,000 troy ounces of silver at the futures price of USD 16.96 per ounce. According to NYMEX rules, the initial margin of USD 6,400 and a maintenance margin of USD 3,000 is required to enter and retain the futures position.

If the price of silver futures contracts dropped to USD 16.92, USD 16.90, and USD 16.73, at the end of the first, second and third day, respectively, then what is the margin account balance at the end of the second day?

- A. USD 5,750
- B. USD 1,850
- C. USD 6,100
- D. USD 2,700

Correct answer: C

Explanation: At the end of the first day, when the price of silver futures contract dropped from USD 16.96 to USD 16.92 per ounce, the margin account balance declined by:

$$(\text{USD } 16.96 - \text{USD } 16.92) \times 5000 = \text{USD } 200$$

At the end of first day, the margin account balance = USD 6,400 - USD 200 = USD 6,200.

At the end of the second day, when the price of futures contract dropped from USD 16.92 to USD 16.90 per ounce, the margin account balance declined by  $(\text{USD } 16.92 - \text{USD } 16.90) \times 5000 = \text{USD } 100$

At the end of second day, the margin account balance = USD 6,200 - USD 100 = USD 6,100

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 7: Futures Markets, Learning objective: Describe the rationale for margin requirements and explain how they work**

47. Christina Forst, an investment manager, has constructed a portfolio that somewhat mirrors the Nasdaq composite index. The investment manager intends to hedge the portfolio by taking a short position in Nasdaq futures. The current worth of the portfolio is USD 633 million, and the Nasdaq index futures price is 2,906, with each contract on 250 times the index. If the beta of the portfolio is 0.98, then the number of contracts Lange should short to hedge her portfolio is closest to:

- A. 871 futures contracts
- B. 854 futures contracts
- C. 217,825 futures contracts
- D. 213,469 futures contracts

Correct answer: B

Explanation: The number of stock contracts required to hedge the portfolio is calculated as:

$$\text{Number of stock contracts} = \text{Beta of the portfolio} \times \frac{\text{Value of the portfolio}}{\text{Futures price}} \times \text{Contract multiplier}$$

Since the portfolio doesn't perfectly mirror the Nasdaq composite index, the beta of the portfolio of 0.98 will be considered in the calculation.

$$\text{The number of contracts required} = 0.98 \times \left( \frac{633,000,000}{726,500} \right) = 853.8$$

Therefore, the number of contracts required is closest to 854.

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 6: Using Futures for Hedging, Learning objective: Compute the optimal number of futures contracts needed to hedge an exposure and explain and calculate the "tailing the hedge" adjustment.**

48. You have been provided the following table of zero spot rates:

| Year | Zero rates (p.a.) |
|------|-------------------|
| 1    | 5.6%              |
| 2    | 6.0%              |
| 3    | 8.4%              |
| 4    | 9.8%              |

Using the information provided in the table, which of the following is the accurate 1-year forward rate 2 years from now?

- A. 12.6%
- B. 15.6%
- C. 10.8%
- D. 13.2%

Correct answer: D

Explanation: The 1-year forward rate 2 years from now is calculated as:

$$\text{Forward rate} = \text{Spot rate}_{\text{year 3}} + (\text{Spot rate}_{\text{year 3}} - \text{Spot rate}_{\text{year 2}}) \times \frac{\text{Time 1}}{\text{Time 2} - \text{Time 1}}$$

$$\text{Forward rate} = 0.084 + (0.084 - 0.060) \times \frac{2}{3 - 2} = 0.132$$

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 16: Properties of Interest Rates, Learning objective: Derive forward interest rates from a set of spot rates.**

49. Blake Young is a derivative investment manager at Yankee Bank based in New York City. He manages a portfolio of fixed income assets and interest rate futures. He currently has a short position in a futures contract on GILTS, the U.K. equivalent to U.S. Treasury securities. As the delivery month is approaching, Young's manager has to choose the cheapest-to-deliver bond from the four available bonds.

If the last settlement price is 93-08, which of the following bond is the cheapest-to-deliver?

| Bond | Quoted Bond Price | Conversion Factor |
|------|-------------------|-------------------|
| A    | USD 102           | 1.018             |
| B    | USD 103           | 1.071             |
| C    | USD 103           | 1.089             |
| D    | USD 104.50        | 1.062             |

- A. Bond A
- B. Bond B
- C. Bond C
- D. Bond D

Correct answer: C

Explanation: The cheapest-to-deliver bond among the four available bonds is Bond C.

Settlement price of 93-08 = 93.25

To find out cheapest-to-deliver bond we will use the following formula:

Cost of delivery = Quoted bond price – (Last settlement price × Conversion factor)

| Bond | Quoted Bond Price | Conversion Factor | Cost of Delivery                                   |
|------|-------------------|-------------------|----------------------------------------------------|
| A    | USD 102           | 1.018             | $102 - (93.25 \times 1.018) = \text{USD } 7.07$    |
| B    | USD 103           | 1.071             | $103 - (93.25 \times 1.071) = \text{USD } 3.13$    |
| C    | USD 103           | 1.089             | $103 - (93.25 \times 1.089) = \text{USD } 1.45$    |
| D    | USD 104.50        | 1.062             | $104.50 - (93.25 \times 1.062) = \text{USD } 5.47$ |

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 19: Interest Rates Futures, Learning objective: Calculate the cost of delivering a bond into a Treasury bond futures contract.**



50. A retail trader is considering borrowing 1000 Euros and investing the proceeds in 1.50-year currency futures contracts on the U.S. dollar. The 1.5-year risk-free interest rates in the Eurozone and the U.S. are 1.36% and 1.55%, respectively, and the spot exchange rate is 1.071 USD per Euro. According to the given data, the futures exchange rate should be 1.074 USD per Euro. However, the 1.5-year futures exchange rate is quoted at 1.100 USD per Euro.

Using this information outlined above, which of the following best represents the arbitrage gain or loss from the strategy?

- A. Arbitrage gain of USD 26.48
- B. Arbitrage loss of USD 26.48
- C. Arbitrage gain of USD 2.06
- D. Arbitrage loss of USD 2.06

Correct answer: B

Explanation: The investor borrows €1000 for 1.5 years at the rate of 1.36% and converts the €1000 into USD at the current exchange rate:

$$€1000 \times 1.071 = \text{USD } 1071$$

Then, the investor will invest the USD 1071 at the U.S. risk-free rate of 1.55% continuously compounded for 1.5 years:

$$\text{USD } 1071 \times e^{(0.0155 \times 1.5)} = \text{USD } 1,096.19$$

$$\text{After 1.5 years, the euro loan payable will equal } €1000 \times e^{(0.0136 \times 1.5)} = €1,020.61$$

The futures contract to buy €1,020.61 at the futures exchange rate of 1.100 USD per EUR costs  $€1,020.61 \times 1.100 = \text{USD } 1,122.67$

The risk-free gain/loss at the end of 1.5 years is  $\text{USD } 1,096.19 - \text{USD } 1,122.67 = - \text{USD } 26.48$

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 8: Commodity Forwards and Futures, Learning objective: Describe an arbitrage transaction in commodity forwards and compute the potential arbitrage profit**

51. Tiara Enterprises (TIEN) has just announced its plans to establish a facility in New York, USA, to meet the increased demand for its products. TIEN plans to fund the expansion with debt, and in order to hedge the risk of borrowing, TIEN has entered into a plain vanilla interest rate swap with a notional principal of USD 50 million. TIEN would make semiannual payments at the rate of 12% with the counterparty making floating rate payments at the Euribor rate.

Assuming a 360-day year, if the Euribor was 13.5% on the last settlement date and is 11.0% on the current settlement date, the amount that TIEN would receive at the current settlement date is *closest to*:

- A. USD 250,000.
- B. USD 625,000.
- C. USD 375,000.
- D. USD 312,500

Correct answer: C

Explanation: TIEN's payment:  $USD\ 50\ million \times \frac{180}{360} \times 12\% = USD\ 3,000,000$

Counterparty's payment:  $USD\ 50\ million \times \frac{180}{360} \times 13.5\% = USD\ 3,375,000$

So TIEN would receive: USD 375,000 (net)

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 20: SWAPS, Learning objectives:**

- *Calculate the value of a plain vanilla interest rate swap based on two simultaneous bond positions.*
- *Calculate the value of a plain vanilla interest rate swap from a sequence of FRAs.*

52. James Porter is analyzing in the following option contracts on a security with a current market price of USD 20:

**Exhibit: Option Contracts**

| Call Options | Exercise Price | Time to Expiration |
|--------------|----------------|--------------------|
| A            | USD 25         | 36 months          |
| B            | USD 29         | 8 months           |
| C            | USD 30         | 22 months          |

Which of the above option contracts is *more likely* to have the lowest premium?

- A. Call A
- B. Call B
- C. Call C
- D. Either Call A or Call B

Correct answer: B

Explanation: Call option B has a higher exercise price than option A and a lower time to expiration. Even though the exercise price is lower than that of Option C, Option C has a much longer time to expiration, and hence should be more expensive.

***FRM Part 1, Book 3: Financial Markets and Products, Chapter 13: Properties of Options, Learning objective: Identify the six factors that affect an option's price***

53. A fund manager has realized that there is a great potential for profits in the options market without tying up much capital. To test the potential of options trading, he implemented a spread strategy by purchasing a 6-month European call with the strike price of USD 92 and a 6-month European put option with a strike price of USD 84 on the stock of XTR.

Which of the following strategies is the fund manager most likely using?

- A. Straddle strategy
- B. Strip strategy
- C. Strap strategy
- D. Strangle strategy

Correct answer: D

Explanation: A strangle trading strategy is implemented by buying a European call and a European put option on the stock of a specific firm with the same expiration date but different strike prices.

Option A is incorrect because, in a straddle strategy, the investor buys a European call option and put options with the same strike price and expiration.

Option B is incorrect because a strip options trading strategy is implemented by purchasing a European call option and purchasing two European put options with the same strike price and expiry date.

Option C is incorrect. In a straps trading strategy, the investor purchase two European call options and one European put option with the same strike price and expiration date.

***FRM Part 1, Book 3: Financial Markets and Products, Chapter 14: Trading Strategies, Learning objectives:***

- *Explain the motivation to initiate a covered call or a protective put strategy.*
- *Describe the use and explain the payoff functions of combination strategies.*

54. A trader is interested in a strategy in which she purchases an option with the strike price of USD 92 that expires in April. She then invests in another option that starts in April and expires in July. This option will have a strike price equal to the price of the underlying in April. The series of such options is called a:

- A. Cliquet option
- B. Gap option
- C. Forward start option
- D. Spiral futures option

Correct answer: A

Explanation: A cliquet option is an exotic option consisting of a series of consecutive forward start options. The first option is active immediately. The second then becomes active when the first expires, etc. Each option is struck at-the-money when it becomes active.

Option B is incorrect. A gap option is a type of binary option whose stated strike price is different from its payoff strike.

Option C is incorrect. A forward start option starts at a specified date in the future; however, the premium is paid in advance, and the time of expiration is established at the time the forward start option is purchased.

Option D is incorrect. A spiral futures option is not a type of option.

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 14: Exotic Options, Learning objective: Identify and describe the characteristics and payoff structure of the following exotic options: gap, forward start, compound, chooser, barrier, binary, lookback, Asian, exchange, and basket option.**

55. Jon Pham, a commodities trader, has analyzed the forward prices of crude oil contracts and realized that there might be an arbitrage profit present in oil futures contracts. The spot price for one barrel of crude oil is USD 62.57, and the 6-month futures contract is quoted as USD 65.35 per barrel. If the risk-free rate is 5%, then the arbitrage profit for trading one crude oil futures contract is closest to:

- A. USD 0
- B. USD 1.20
- C. USD 0.35
- D. USD 0.42

Correct answer: B

Explanation: The futures price of the 6-month crude oil contract is greater than the spot price. Therefore, an arbitrage opportunity exists.

An investor should take the following steps at time 0:

1. Borrow USD 62.57 for 6 months at the risk-free rate
2. Buy one barrel of crude oil at the spot price of USD 62.57
3. Take a short exposure in the 6-month crude oil contract for the futures price of USD 65.35

At the expiration of the futures contract:

1. Sell crude oil at the futures price of USD 65.35
2. Pay the borrowed money with interest, i.e.,  $\text{USD } 62.57 \times e^{0.05 \times 0.5} = \text{USD } 64.15$
3. Earn the cash-and-carry arbitrage profit of  $(\text{USD } 65.35 - \text{USD } 64.15) = \text{USD } 1.20$  per barrel

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 11: Commodity Forwards and Futures, Learning objective: Describe an arbitrage transaction in commodity forwards and compute the potential arbitrage profit**

56. Barbara Stapleton has recently joined Ace Investments as an investment manager. Her new boss told her that she would only be trading derivatives on exchanges, and the firm does not approve the use of over-the-counter derivatives. Which of the following derivative instruments is she allowed to trade?

I. Forwards

II. Options

III. Swaps

IV. Futures

- A. I and II only
- B. II and IV only
- C. I, II and IV only
- D. All of the above

Correct answer: B

Explanation: If the investment company does not approve the trading of over-the-counter derivatives, then the trading of forwards and swaps is not allowed. Generally, forwards and swaps trade on over-the-counter markets while futures contracts and options trade on exchanges.

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 5: Exchanges and OTC Markets, Learning objective: Compare exchange-traded and OTC markets and describe their uses.**

57. Which of the following statements is (are) incorrect regarding central counterparties (CCP)?

- I. The margining activity of the central counterparty decreases risk, but in some cases, it can also increase risk
  - II. The central counterparty does not make counterparty risk disappear; rather, it centralizes and converts counterparty risk into different forms of financial risk
  - III. Unlike other financial institutions, the central counterparty cannot fail
- A. I only
  - B. III only
  - C. I and II
  - D. I and III

Correct answer: B

Explanation: Option III is incorrect because, like any other financial institutions, the central counterparty can also fail.

Option I is correct. The increase of margin requirements can absorb the liquidity of the party, which can result in increased counterparty risk or default risk.

Option II is correct. It is true that the central party doesn't eliminate counterparty risk completely, but, in fact, it centralizes the risk by acting as the counterparty to every party.

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 6: Central Clearing, Learning objectives:**

- **Provide examples of the mechanics of a central counterparty (CCP).**
- **Describe the role of CCPs and distinguish between bilateral and centralized clearing.**
- **Describe the advantages and disadvantages of central clearing of OTC derivatives**

58. Which of the following is the main source of liquidity risk in central counterparties?

- A. High operating costs
- B. Overinvestment in long-term assets
- C. Varying margin payments
- D. Overinvestment in derivatives assets

Correct answer: C

Explanation: The main reason why CCPs face liquidity risk has much to do with varying margin payments. These contractual payments keep on changing as positions change. If the CCP does not set aside enough funds to meet margin payments, it risks failing to fulfill its obligations to surviving members in a timely manner.

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 6: Central Clearing, Learning objective: Identify and explain the types of risks faced by CCPs.**

59. A foreign currency analyst based in Canada is forecasting the forward exchange rate CAD/USD. Currently, the spot exchange rate is CAD 1.21 per USD. Suppose that the interest rate in Canada is 1%, and the interest rate in the U.S. is 3%, which of the following is closest to the forward exchange rate of CAD per USD?

- A. CAD 1.1865 per USD
- B. CAD 1.2463 per USD
- C. CAD 1.2339 per USD
- D. CAD 1.2221 per USD

Correct answer: A

Explanation: According to interest rate parity theorem, the forward exchange must be derived with the following equation:

$$\begin{aligned}\text{Forward exchange rate of CAD per USD} &= \text{Spot exchange rate} \times \frac{1 + \text{CAD interest rate}}{1 + \text{USD interest rate}} \\ &= \text{CAD } 1.21 \times \left( \frac{1.01}{1.03} \right) = \text{CAD } 1.1865/\text{USD}\end{aligned}$$

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 9: Foreign Exchange Market, Learning objective: Calculate and explain the effect of an appreciation/depreciation of a currency relative to a foreign currency.**

60. Consider the following information:

| Currency | Libor (annualized) | Currency Combinations | Spot Rate |
|----------|--------------------|-----------------------|-----------|
| USD      | 0.30%              | USD/EUR               | 1.6975    |
| EUR      | 5.00%              | JPY/EUR               | 0.0085    |
| JPY      | 0.30%              | JPY/USD               | 82.25     |

If the uncovered interest rate parity holds, what is the forward rate of JPY/EUR currency for one year?

- A. 0.0045
- B. 0.0089
- C. 0.0056
- D. 0.0065

Correct answer: B

Explanation: Since we have assumed that the uncovered interest rate parity holds, then the forward rate parity holds. That is, a one-year spot rate should be equal to the one-year forward rate. That is,

$$F_{\text{JPY/EUR}} = S_{\text{JPY/EUR}} \left( \frac{1 + i_{\text{EUR}}}{1 + i_{\text{JPY}}} \right) = 0.0085 \left( \frac{1.05}{1.003} \right) = 0.0089$$

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 9: Foreign Exchange Market, Learning objective:**

- *Explain how no-arbitrage assumption in the foreign exchange markets leads to the interest rate parity theorem and use this theorem to calculate forward foreign exchange rates.*
- *Distinguish between covered and uncovered interest rate parity conditions.*

61. Joachim Corporation issued four types of debt to raise the capital of USD 730 million. The debt includes USD 400 million raised through secured bonds, USD 110 million through subordinate bonds, USD 40 million through unsecured loans, and USD 180 million through debenture bonds. If the company defaults, the total value of the assets to be distributed would be USD 630 million.

The total claim that the subordinate bondholders would receive is closest to:

- A. USD 0
- B. USD 180 million
- C. USD 50 million
- D. USD 10 million

Correct answer: D

Explanation: Owners of subordinated debt stand last in line among creditors when an issuer fails financially.

Since the total value of the firm's asset after default would be USD 630 million:

The secured bondholders will receive USD 400 million

The debenture bondholders will receive USD 180 million

The banks will receive USD 40 million

The subordinate bondholders will receive USD 10 million

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 17: Corporate Bonds, Learning objectives:**

- *Describe the different classifications of bonds characterized by the issuer, maturity, interest rate, and collateral.*
- *Evaluate the expected return from a bond investment and identify the components of the bond's expected return.*



62. In a planned amortization class (PAC) collateralized mortgage obligation (CMO), when compared to the underlying mortgage-backed security, the planned amortization class (PAC) tranches have:

- A. Reduced contraction risk
- B. Reduced extension risk
- C. Both reduced contraction risk and reduced extension risk
- D. None of the above

Correct answer: C

Explanation: A PAC tranche is structured to make predictable payments, regardless of actual prepayments to the underlying MBS. The PAC tranches have both reduced contraction risk and reduced extension risk compared to the underlying MBS.

***FRM Part 1, Book 3: Financial Markets and Products, Chapter 18: Mortgages and Mortgages-backed Securities, Learning objective: Explain the mechanics of different types of agency MBS products, including collateralized mortgage obligations (CMOs), interest-only securities (IOs) and principal-only securities (POs)***

63. Which of the following statements is incorrect regarding mortgage-back securities?

- A. Commercial mortgage-backed securities tend to be less volatile and complex compared to residential mortgage-backed securities
- B. The risk of prepayment typically occurs in declining rate environments
- C. When you invest in a mortgage-backed security, you are indirectly lending money to a home buyer or business
- D. None of the above

Correct answer: A

Explanation: Commercial mortgage-backed securities are a type of mortgage-backed security backed by commercial mortgages rather than residential real estate and tend to be more complex and volatile than residential mortgage-backed securities due to the unique nature of the underlying property assets.

***FRM Part 1, Book 3: Financial Markets and Products, Chapter 18: Mortgages and Mortgages-backed Securities, Learning objectives:***

- ***Describe the various types of residential mortgage products.***
- ***Describe the mortgage prepayment option and the factors that influence prepayments. Describe the process of trading of pass-through agency MBS.***

64. What is the most likely definition of a Special Purpose Vehicle (SPV)?

- A. A subsidiary company with an asset/liability structure and legal status that makes its obligations secure even if the parent company goes bankrupt
- B. A type of asset-backed security that is secured by a mortgage or collection of mortgages
- C. A structured financial product that pools together cash flow-generating assets and repackages this asset pool into discrete tranches that can be sold to investors
- D. A subsidiary that reduces the effects of prepayment risk – the risk that the principal amount will be paid ahead of schedule

Correct answer: A

Explanation: A Special Purpose Vehicle (SPV) is a subsidiary of a company which is bankruptcy-remote from the main organization. The actions of an SPV are usually very tightly controlled, and they are only allowed to finance, buy, and sell assets.

Option B is the definition of mortgage-backed security.

Option C is the definition of a CDO.

Option D is partially the definition of a PAC tranche. However, a PAC tranche is not a subsidiary.

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 5: Exchanges and OTC Markets, Learning objective: Explain the use of special purpose vehicles (SPVs) in the OTC derivatives market.**

65. Which of the following statements is correct regarding the value of a forward contract to a short party at expiration?

The value of the forward contract is:

- A. Equal to zero
- B. Equal to the value of the long party multiplied by -1
- C. Positive if the spot price of the underlying exceeds the forward price
- D. Either the full amount of the contract or zero

Correct answer: B

Explanation: The value of the forward contract to a party holding a short position can be calculated by multiplying the value to the long party by -1.

Option A is incorrect. The forward contract most likely has a value at expiration, and this value is equal to the difference between the forward price and the underlying current spot price.

Option C is incorrect. The value of the forward contract to the party with a short position is positive if the futures price exceeds the spot price. This party, with a short position, has agreed to deliver the obligation for a price, which is higher than what would have been received if it were sold today in the market.

Option D is incorrect. The value of the forward contract is not either the full amount of the contract or zero; rather, it is the value of the long position multiplied by -1.

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 15: Commodity Forwards and Future, Learning objectives:**

- ***Explain how to create a synthetic commodity position and use it to explain the relationship between the forward price and the expected future spot price.***
- ***Explain the relationship between current futures prices and expected future spot prices, including the impact of systematic and nonsystematic risk.***

66. Jason Briggs purchased a 3-month call option by paying USD 0.08. The exercise price of the option is USD 1.32, while the underlying is priced at USD 1.35.

Is the option currently in-the-money, and at what price will break-even occur?

- A. In-the-money: No; Break-even Price: USD 1.27.
- B. In-the-money: Yes; Break-even Price: USD 1.40.
- C. In-the-money: Yes; Break-even Price: USD 1.35.
- D. In-the-money: No; Break-even Price: USD 1.35.

Correct answer: B

Explanation: The call option is in-the-money because the underlying price is greater than the exercise price (USD 1.35 vs. USD 1.32, respectively).

Break-even price =  $X + C_0$  = USD 1.32 + USD 0.08 = USD 1.40

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 11: Options Markets, Learning objective: Describe the types, position variations, payoffs and profits and typical underlying assets of options.**

67. A European put option is selling for USD 4.00 with the underlying priced at USD 52. The exercise price is USD 50, and the underlying makes no cash payments during the life of the option. The risk-free rate is 6.0%, and the option expires in 120 days. A call with the same exercise price and expiry sells for USD 8.50.

This put option is most likely:

- A. Overvalued by USD 1.55
- B. Undervalued by USD 1.55
- C. Overvalued by USD 3.67
- D. Undervalued by USD 3.67

Correct answer: B

Explanation:

$$P_0 = C_0 - S_0 + \frac{X}{(1+r)^T}$$
$$= 8.50 - 52 + \frac{50}{(1.06)^{\frac{120}{365}}} = 5.55$$

Hence, the put at the price of USD 4 is undervalued by USD 5.55 – USD 4 = USD 1.55.

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 11: Properties of Options, Learning objective: Explain put-call parity and apply it to the valuation of European and American stock options, with dividends and without dividends and express it in terms of forward prices.**

68. A trader applied a strategy where he purchased a European call option on the stock of KKL with a strike price of USD 29.50, and at the same time, sold a European call option on the same stock with a strike price of USD 34.89. Suppose that the final price of the stocks at expiration is USD 44, what are the name and the payoff of the strategy? Ignore the cost of the strategy.

- A. Name of the strategy: Bear call spread; Payoff: USD -9.11
- B. Name of the strategy: Bull call spread; Payoff: USD 9.11
- C. Name of the strategy: Bear call spread; Payoff: USD -5.39
- D. Name of the strategy: Bull call spread; Payoff: USD 5.39

Correct answer: D

Explanation: In a bull spread strategy, an investor buys European call options with a specific strike price (USD 29.50) and simultaneously sells European call options with a higher strike price (USD 34.89).

If the current price (USD 44) is higher than the strike price of the short call option (USD 32), both call options will be exercised, and the payoff of the investor will be  $X_2 - X_1$  or USD 34.89 - USD 29.50 = USD 5.39.

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 14: Trading Strategies, Learning objectives:**

- Describe the use and calculate the payoffs of various spread strategies.
- Describe the use and explain the payoff functions of combination strategies

69. Janet Marshall, a fixed investment manager at Premium S&B, has a short position in 10-year Treasury bond contracts with a USD 100,000 face value for each contract. The last quoted price of the contract is 98-30, while the accrued interest on the bond is USD 5.23 (for USD 100 face value). If the conversion factor for the deliverable bond under the contract is 1.375, then the cash price:

- A. Paid by the short party is USD 140.393
- B. Received by the short party is USD 140.393
- C. Paid by the short party is USD 141.269
- D. Received by the short party is USD 141.269

Correct answer: D

Explanation: Janet Marshall has a short position in the contract. Therefore, the applicable quoted price derived by the conversion factor is the price Marshall will receive.

The cash price received by the short party of the contract is calculated as:

$$\begin{aligned}\text{Cash received by the short party} &= (\text{Most recent settlement price} \times \text{Conversion factor}) + \text{Accrued interest} \\ &= (98 + 30/32) \times 1.375 + \text{USD } 5.23 = \text{USD } 141.269\end{aligned}$$

or USD 141.269 for a USD 100,000 face value contract

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 19: Interest Rate Futures, Learning objective: Calculate the theoretical futures price for a Treasury bond futures contract**

70. A deliverable bond has a maturity of 12 years and 7 months and pays a 6% coupon every six months. What is the conversion factor of the bond given that the discount rate is 7%, and the face value of the bond is USD 100 (maturity rounded to the nearest 3 months)?

- A. 0.9176
- B. 0.9172
- C. 0.9185
- D. None of the above

Correct answer: A

Explanation:

$$\text{Conversion factor} = \frac{(\text{Discount price} - \text{Accrued interest})}{\text{Face value}}$$

Usually, exchanges round up (or down) the maturity years to the nearest 3 months so the 12 years and 7 months maturity will be rounded down to 12 years and 6 months. Therefore,  $N = 12.5 \times 2 = 25$

The discounted value of the bond can also be calculated using a financial calculator with the following inputs:

$N=25$ ,  $I/Y=3.5$ ,  $FV=100$ ,  $PMT=3$ ,  $CPT \rightarrow PV = 91.7592$

$$\text{Conversion factor} = \frac{91.7592}{100} = 0.9176$$

**FRM Part 1, Book 3: Financial Markets and Products, Chapter 9: Interest Rate Futures, Learning objective: Explain and calculate a US Treasury bond futures contract conversion factor**

### Valuation and Risk Models

71. Anna Frost is a junior options trader at an investment bank. She recently finalized a non-standardized options contract with a new client. The option had the following Greeks:

|              |     |
|--------------|-----|
| <b>Delta</b> | 25  |
| <b>Gamma</b> | 100 |
| <b>Vega</b>  | 100 |

Although Frost's supervisor does not mind bearing the risk of the changes in the price of the underlying, he is not comfortable with the Gamma and Vega risk exposures and asks Frost to hedge her position. At the moment, she has the following two options available on the market:

|              | <b>Call Option 1</b> | <b>Call Option 2</b> |
|--------------|----------------------|----------------------|
| <b>Delta</b> | 0.55                 | 0.85                 |
| <b>Gamma</b> | 0.25                 | 0.1                  |
| <b>Vega</b>  | 30                   | 10                   |

Which trades should Anna undertake to accomplish her supervisor's task?

- A. Buy 2,131 contracts of Call Option 1 and sell 5,639 contracts of Call Option 2
- B. Sell 1,980 contracts of Call Option 1 and buy 2,131 contracts of Call Option 2
- C. Buy 5,950 contracts of Call Option 1 and buy 1,980 contracts of Call Option 2
- D. Buy 1,980 contracts of Call Option 1 and sell 5,950 contracts of Call Option 2

Correct answer: D

Explanation: Let's denote  $a$  as the number of the transactions in Call Option 1 and  $b$  the number of the transactions in Call Option 2.

$$\begin{cases} 100 + 0.25 \times a + 0.1 \times b = 0 \\ 100 + 30 \times a + 10 \times b = 0 \end{cases}$$

$$0.25a + 0.1b = 30a + 10b$$

$$-29.75a = 9.9b$$

$$a = -0.3328b$$

$$100 + 30(-0.3328b) + 10b = 0$$

$$100 + 0.016b = 0$$

$$b = -5,950$$

$$a = 1,980 \text{ and } b = -5,950$$

Buying 1,980 contracts of Call Option 1 and selling 5,950 contracts of Call Option 2 will make the portfolio gamma and vega neutral.

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 16: Option Sensitivity Measures: The Greek Letters, Learning objectives:**

- **Define and describe theta, gamma, vega, and rho for options positions and calculate the gamma and vega for a portfolio.**
- **Explain how to implement and maintain a delta-neutral and a gamma-neutral position.**

72. An investment bank decided to open a position in the shares of ALNY's stock. The position will be opened by buying 5,000 out-of-the-money call options with a strike price of USD 350. At the moment, the stock is trading at USD 300. The annual volatility of the stock's return is 30%, and the delta of the option is 0.3. What is the daily VaR for this position at a 1% significance level, assuming 252 trading days in a year?

- A. USD 16,668
- B. USD 19,815
- C. USD 23,116
- D. USD 16,282

Correct answer: B

Explanation: The one-tail z value for a confidence interval of 99% is 2.33.

The VaR of position =  $\frac{30\%}{\sqrt{252}} \times 2.33 \times \text{USD } 300 \times 0.3 \times 5000 = \text{USD } 19,815$  (to 1 d.p)

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 2: Calculating and Applying VaR, Learning objective: Describe and calculate VaR for linear derivatives**



73. Ganz Hertz is the CEO of one of the biggest investment funds in Germany. Although the fund has always followed a balanced equity/fixed income allocation strategy, a recent rally in the real estate market prompted a reevaluation. Fritz Wolfgang, the chief investment officer, proposed to enlarge the fund's investment horizon and reallocate 10% of the fund's assets from bonds to real estate.

According to Wolfgang, the investment in a new asset class, which is not perfectly correlated with stocks and bonds, will not only help to increase returns but will also diversify risks and decrease the firm's VaR. To prove that Wolfgang is incorrect, Hertz quickly calculates VaR for the stock, bond, and real estate portfolios separately add them up (taking into account the proposed allocation proportions), and compares the final result with the VaR of the current portfolio. Wolfgang is disappointed with results because the VaR of the current portfolio turns out to be lower than the one calculated by Hertz.

Based on the information provided, which of the following statements is correct?

- A. The impact of the new asset class inclusion on portfolio VaR is correct, and the portfolio VaR calculation approach is also correct
- B. The impact of the new asset class inclusion on portfolio VaR is correct, while the portfolio VaR calculation approach is incorrect
- C. The impact of the new asset class inclusion on portfolio VaR is incorrect, while the portfolio VaR calculation approach is correct
- D. The impact of the new asset class inclusion on portfolio VaR is incorrect, and the portfolio VaR calculation approach is also incorrect

Correct answer: D

Explanation: Non-perfect correlation does not necessarily mean a reduction in portfolio VaR because the impact will also depend on the variance of the new asset class.

The approach taken by Hertz is incorrect as it fails to take into account the non-perfect correlations between stocks, bonds, and real estate.

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 1: Measures of Financial Risk, Learning objective: Define the VaR measure of risk, describe assumptions about return distributions and holding period, and explain the limitations of VaR.**

74. Which of the following statements is (are) accurate?

- I. VaR is not a coherent risk measure while Expected Shortfall is
  - II. Although the calculation of Expected Shortfall is more difficult, it provides a narrower range of expected losses than VaR does under the same significance level
- A. Both I and II
  - B. I only
  - C. II only
  - D. None of the above

Correct answer: B

Explanation: Statement I is correct. Expected Shortfall is a coherent risk measure, while VaR is not because it fails the subadditivity criteria.

Statement II is incorrect. Expected Shortfall provides the value of expected losses and not the range.

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 1: Measures of Financial Risk, Learning objectives:**

- ***Explain and calculate ES and compare and contrast VaR and ES.***
- ***Define the properties of a coherent risk measure and explain the meaning of each property.***
- ***Explain why VaR is not a coherent risk measure.***

75. Donat Bank from Slovenia recently started trading in FX options. Senior management wants to receive reports about risks arising from those trading activities and wants from its risk management unit explanations regarding the techniques used to measure risk. Which technique(s) is (are) most appropriate in this case?

- A. Delta normal
- B. Full revaluation
- C. Both A and B
- D. None of the above

Correct answer: C

Explanation: There are two primary approaches to the measurement of the risk of nonlinear securities. The first is the most straightforward approach – the full revaluation approach. This approach is predicated on the fact that the derivative moves one-for-one, or one-for-delta, with the underlying factor.

The alternative is the approach known as the “delta-normal” approach, which involves the delta (linear) approximation, or the delta-gamma (Taylor Series) approximation.

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 2: Calculating and Applying VaR, Learning objectives:**

- ***Describe and calculate VaR for linear derivatives.***
- ***Describe and explain the historical simulation approach for computing VaR and ES.***
- ***Describe the delta-normal approach for calculating VaR for non-linear derivatives.***
- ***Explain the full revaluation method for computing VaR.***
- ***Compare delta-normal and full revaluation approaches for computing VaR.***

76. Because of the recent volatility of interest rates, the CRO of a hedge fund proposes to completely eliminate the 5-year, 10-year, and 30-year exposure of the fixed income portfolio. The table below presents the key rate '01s of the current portfolio and corresponding hedging instruments:

| Hedging Bonds          | Key Rate '01 (per 100 face amount) |          |          |
|------------------------|------------------------------------|----------|----------|
|                        | 5-year                             | 10-year  | 30-year  |
| Bond 1                 | 0.0175                             | -        | -        |
| Bond 2                 | 0.0099                             | 0.0250   | -        |
| Bond 3                 | 0.0232                             | 0.0314   | 0.0487   |
|                        | Key Rate '01 (USD )                |          |          |
|                        | 5-year                             | 10-year  | 30-year  |
| Fixed Income Portfolio | 80.4895                            | 193.4292 | 300.0000 |

What is the CRO's hedging transaction?

- A. Sell USD 357,000 of Bond 1; Sell USD 127,000 of Bond 2; Sell USD 616,000 of Bond 3
- B. Buy USD 270,000 of Bond 1; Sell USD 127,000 of Bond 2; Buy USD 616,000 of Bond 3
- C. Sell USD 270,000 of Bond 1; Buy USD 127,000 of Bond 2; Sell USD 616,000 of Bond 3
- D. Buy USD 357,000 of Bond 1; Sell USD 616,000 of Bond 3

Correct answer: D

**Explanation:** Let's denote  $X_i$  as the face amount of hedging bond  $i$  that we need to sell.

We have following equation

$$(1): X_1 \times \frac{0.0175}{100} + X_2 \times \frac{0.0099}{100} + X_3 \times \frac{0.0232}{100} + + = USD 80.4895$$

$$\Rightarrow 0.0175X_1 + 0.0099X_2 + 0.0232X_3 = USD 8048.95$$

$$(2): X_2 \times \frac{0.0250}{100} + X_3 \times \frac{0.0314}{100} = USD 193.4292$$

$$\Rightarrow 0.0250X_2 + 0.0314X_3 = USD 19342.92$$

$$(3): X_3 \times \frac{0.0487}{100} = USD 300$$

$$\Rightarrow 0.0487X_3 = USD 300$$

$$\text{From (3): } X_3 = USD 616,016.4271$$

$$\text{From (2) and (3): } X_2 = USD 0$$

$$\text{From (1), (2) and (3): } X_1 = -USD 356,721.7776$$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 14: Modeling and Hedging Non-Parallel Term Structure Shifts, Learning objective: Calculate the key rate exposures for a given security and compute the appropriate hedging positions given a specific key rate exposure profile.**

77. The following table provides the initial price of a C-STRIP and its present value after application of a one basis point shift in four key rates.

|               | Value    |
|---------------|----------|
| Initial value | 26.11485 |
| 2-year shift  | 26.11582 |
| 5-year shift  | 26.11778 |
| 10-year shift | 26.13883 |
| 30-year shift | 26.01772 |

The key rate duration with respect to the 30-year shift is closest to:

- A. 35.55
- B. 26.02
- C. 37.19
- D. 97.13

Correct answer: C

Explanation: First, determine the corresponding key rate '01:

$$\text{Key rate '01} = -\frac{\Delta BV}{10,000 \times \Delta y}$$

Where:

$\Delta BV$  = change in bond value

$\Delta y$  = change in yield (0.01%)

The change in bond value here is measured in reference to the initial bond value.

$$= -\frac{26.01772 - 26.11485}{10,000 \times 0.01\%} = 0.09713$$

Recall that:

$$\text{Duration Measure} = \frac{10,000 \times (01\text{Measure})}{\text{Value of the Portfolio}}$$

Therefore in this case, the key rate duration with respect to the 30-year shift can be calculated as:

$$\Rightarrow \text{Key Rate Duration} = \frac{10,000 \times 0.09713}{26.11485} = 37.19$$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 14: Modeling and Hedging Non-Parallel Term Structure Shifts, Learning objective: Define, calculate, and interpret key rate '01 and key rate duration.**

78. Banca Antiniana from Naples, Italy, needs to calculate the risk level of several bond portfolios. Particularly, the bank needs to measure and hedge the risk of portfolios containing both bonds and swaps. What are the instruments which will be used in the calculation?

- A. Partial '01s
- B. Forward-bucket '01s
- C. Both partial '01s and forward-bucket '01s
- D. None of the above

Correct answer: A

Explanation: Key rate exposures are used for measuring and hedging the risk of bond portfolios in terms of a relatively small number of the most liquid bonds available, usually the most recently issued, near-par, government bonds.

Partial '01s are used for measuring and hedging the risk of portfolios of swaps or portfolios that contain both bonds and swaps in terms of the most liquid money market and swap instruments.

Forward-bucket '01s, mostly used in the swap or combined bond and swap contexts, measure the risk of a portfolio not in terms of other securities but in terms of direct changes in the shape of the term structure.

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 14: Modeling and Hedging Non-Parallel Term Structure Shifts**

**Learning objectives:**

- **Calculate the key rate exposures for a given security and compute the appropriate hedging positions given a specific key rate exposure profile.**
- **Relate key rates, partial '01s, and forward-bucket '01s and calculate the forward-bucket '01 for a shift in rates in one or more buckets.**
- **Apply key rate and multi-factor analysis to estimating portfolio volatility**

79. Joshua Manson, a bond trader at Capital Bank, has been asked to calculate the price of U.S. Treasury 4-5/8s of June 30, 2020. Joshua was provided with the following information:

|                             |                         |
|-----------------------------|-------------------------|
| <b>Settlement Date</b>      | May 11, 2017            |
| <b>Coupon payment dates</b> | June 30 and December 30 |
| <b>Yield to maturity</b>    | 5.5%                    |

Note that prices of Treasury bonds are calculated using the ACT/ACT day count convention.

What are the clean and dirty prices of the tranche?

- A. Clean price = 97.5035; Dirty price = 99.1808
- B. Clean price = 99.1808; Dirty price = 97.5035
- C. Clean price = 96.4752; Dirty price = 98.1524
- D. Clean price = 98.1524; Dirty price = 96.4752

Correct answer: A

Explanation: The last coupon payment date is December 30, 2016.

The dirty price of the bond on this date is 97.2484 (can be calculated using calculator with input parameters:  $N = 7$ ,  $I/Y = 2.75$  ( $=5.5/2$ ),  $PMT = 2.3125$  ( $= 4.625/2$ ),  $FV = 100$ )

The next coupon payment date is June 30, 2017. There are 182 days in the current coupon period.

As we already know the price as of December 30, 2016, we can calculate the price on the settlement date as follows:

Dirty price of the bond on May 11, 2017 =  $97.2484 \times (1+2.75\%)^{(132/182)} = 99.1808$

Where 132 = number of days from December 30, 2016, through May 11, 2017

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 9: Pricing Conventions, Discounting, and Arbitrage, Learning objectives:**

- ***Differentiate between “clean” and “dirty” bond pricing and explain the implications of accrued interest with respect to bond pricing.***
- ***Describe the common day-count conventions used in bond pricing.***

80. Marmagne Bank from Bourges, France, needs to calculate the risk level of some bond portfolios. In this stage of the calculation, the Bank does not plan to hedge any of the portfolios but only to closely examine and understand the curve risks of the portfolios in question. What would be the instruments best used for the purpose of understanding the curve risks?

- A. Partial ‘01s
- B. Forward-bucket ‘01s
- C. Both partial ‘01s and forward-bucket ‘01s
- D. None of the above

Correct answer: B

Explanation: Key rate exposures are used for measuring and hedging the risk of bond portfolios in terms of a relatively small number of the most liquid bonds available, usually the most recently issued, near-par, government bonds.

Forward-bucket '01s, mostly used in the swap or combined bond and swap contexts as well, measure the risk of a portfolio not in terms of other securities but in terms of direct changes in the shape of the term structure. As a result, forward-bucket '01s are often the most intuitive way to understand the curve risks of a portfolio, but not the quickest way to see the hedges required to neutralize such risks.

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 14: Modeling and Hedging Non-Parallel Term Structure Shifts**

**Learning objectives:**

- *Calculate the key rate exposures for a given security and compute the appropriate hedging positions given a specific key rate exposure profile.*
- *Relate key rates, partial '01s, and forward-bucket '01s and calculate the forward-bucket '01 for a shift in rates in one or more buckets.*
- *Apply key rate and multi-factor analysis to estimating portfolio volatility*

81. A regional bank recently invested in a new bond-pricing software. To test it, the risk management team priced a bond at 3 different rates. The results are presented below:

| Interest Rate | Price of bond |
|---------------|---------------|
| 2.13%         | USD 100.115   |
| 2.15%         | USD 99.100    |
| 2.17%         | USD 98.250    |

What is the estimate of the bond's convexity?

- A. 104.06
- B. 416.25
- C. 10,406.15
- D. 41,624.62

Correct answer: D

Explanation: Convexity is given by:

$$C = \frac{1}{P} \left[ \frac{P^+ + P^- - 2P}{(\Delta r)^2} \right]$$

Where

$P^+$  - a value of the position when all rates increase by  $\Delta r$

$P^-$  - value of the position corresponding to the decrease of all rates by  $\Delta r$

Note that  $\Delta r$  is given in decimal.

$$= \frac{(100.115 + 98.250 - 2 \times 99.100)}{(99.100 \times 0.0002^2)} = 41,624.62$$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 12: Applying Duration, Convexity, and DV01, Learning objective: Define, compute, and interpret the convexity of a fixed-income security given a change in yield and the resulting change in price.**

82. An employee at a large Wall Street bank is estimating the 5% VaR of a specific asset by using historical simulations to estimate the percentile of returns and using exponentially declining weights on past data. This methodology is most likely an example of:

- A. Parametric VaR estimating
- B. Nonparametric VaR estimating
- C. Implied Volatility VaR estimating
- D. Hybrid VaR estimating

Correct answer: D

Explanation: The hybrid VaR estimating methodology is a combination of parametric and nonparametric methods. The employee's method of estimating VaR by using historical simulations (nonparametric method) to estimate the percentile of the returns and using exponentially declining weights (parametric method) on past data is an example of a hybrid methodology.

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 2: Calculating and Applying VaR, Learning objective: Describe and explain the historical simulation approach for computing VaR and ES**



83. Ben Thompson, FRM, is the senior risk manager at FTK bank. The bank has a significant proportion of its portfolio allocated to AAA Holdings. Thompson is concerned with the effect of a recent 5% drop in prices of AAA Holdings on the firm's portfolio VaR. To estimate the effect, he wants first to calculate the updated volatility of AAA Holdings using the GARCH(1,1) model. Thompson knows that the most recent and the long-term daily volatilities of holding are 2.7% and 2%, respectively. He also estimates the model's parameters as:

$$\alpha = 0.12; \beta = 0.8; \text{ and } \omega = 0.000032$$

What is the updated volatility estimate of AAA Holdings?

- A. 2.8%
- B. 2.9%
- C. 3.0%
- D. 4.0%

Correct answer: C

Explanation:  $\sigma_n^2 = \gamma \times V_L + \alpha \times u_{n-1}^2 + \beta \times \sigma_{n-1}^2 = \omega + \alpha \times u_{n-1}^2 + \beta \times \sigma_{n-1}^2 = 0.000032 + 0.12 \times 0.05 \times 0.05 + 0.8 \times 0.027 \times 0.027 = 0.0009152$

$$\sigma_n = \sqrt{0.0009152} = 0.03$$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 3: Measuring and Monitoring Volatility, Learning objective:**

- **Apply the exponentially weighted moving average (EWMA) approach and the GARCH (1,1) model to estimate volatility.**
- **Explain and apply approaches to estimate long-horizon volatility/VaR and describe the process of mean reversion according to a GARCH (1,1) model.**

84. An analyst just finalized his presentation to the investment committee of an investment bank where he showed his calculation of the price of a 1-year European put option on stocks of one of the local companies using a binomial tree model. He used the following inputs:

|                                      |         |
|--------------------------------------|---------|
| <b>Stock price</b>                   | USD 100 |
| <b>Stock price annual volatility</b> | 30%     |
| <b>Option's strike price</b>         | USD 90  |
| <b>Risk-free rate</b>                | 3%      |

A few hours later, he decided to revise the calculations and found a mistake in the calculation of the stock price annual volatility.

How will the price of the European put option change if the correct annual price volatility is 50% instead of 30%?

- A. +USD 8.80
- B. -USD 8.80
- C. +USD 5.25
- D. -USD 5.25

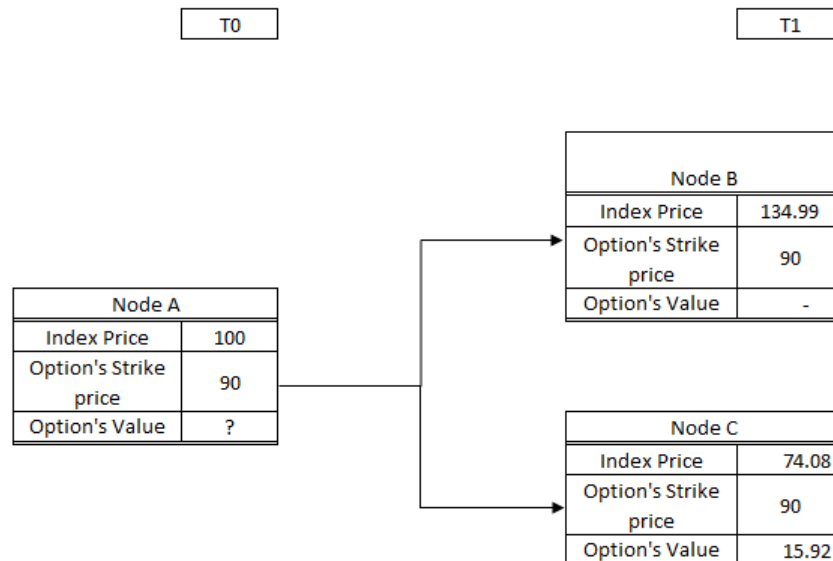
Correct answer: A

Explanation: Using initial inputs:

$$p_{up} = (e^{(r \times \Delta t)} - d) / (u - d)$$

$$= (e^{3\%} - e^{-0.3}) / (e^{0.3} - e^{-0.3}) = 0.4756$$

The figure below shows the stock price and the corresponding option's intrinsic value at each node.

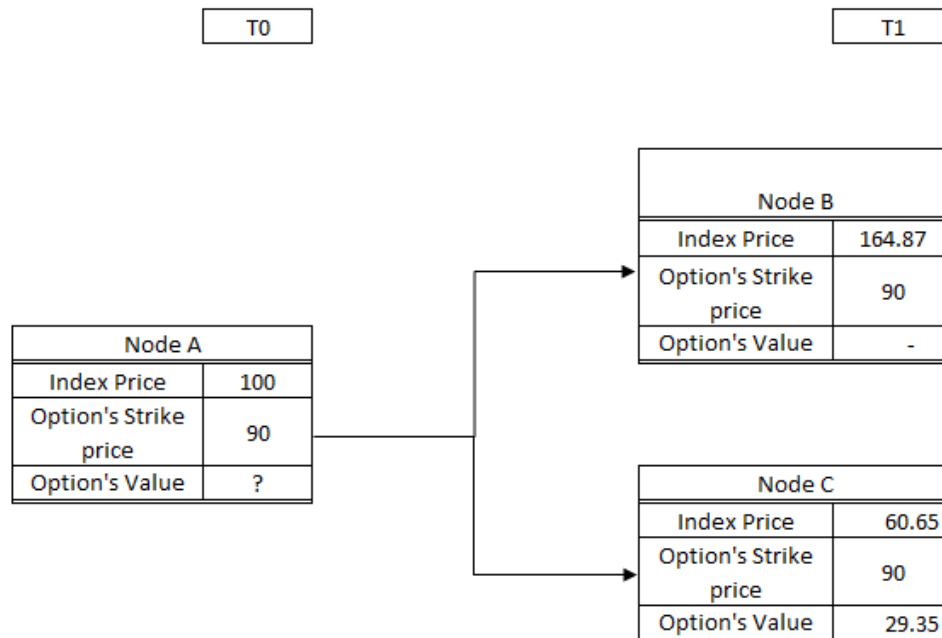


$$\text{Price at A} = (0 \times 0.4756 + 15.92 \times (1 - 0.4756)) \times e^{(-3.0\%)} = 8.10$$

Using corrected inputs:

$$p_{\text{up}} = \frac{e^{r \times \Delta t} - d}{u - d} = \frac{e^{3\%} - e^{-0.5}}{e^{0.5} - e^{-0.5}} = 0.4068$$

The figure below shows the stock price and the corresponding option's intrinsic value at each node.



$$\text{Corrected Price at Node A} = (0 \times 0.4068 + 29.35 \times (1 - 0.4068)) \times e^{-3.0\%} = 16.90$$

The difference in price is  $16.9 - 8.1 = +8.8$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 14: Binomial Trees, Learning objectives:**

- Calculate the value of an American and a European call or put option using a one-step and two-step binomial model.
- Describe how volatility is captured in the binomial model

85. Futures on one of the Fortune 500 companies are correctly trading at USD 88 per contract. The volatility of these futures contracts is 21%. What is the price of a one-year European call option with a USD 85 strike price on this futures contract using the one-step binomial tree model if the risk-free rate is 3% per year?

- A. USD 11.43
- B. USD 11.86
- C. USD 12.21
- D. USD 23.56

Correct answer: B

Explanation:  $p_{up} = \frac{(1+r)-d}{u-d} = \frac{1.03-e^{-0.21}}{e^{0.21}-e^{-0.21}} = 0.5186$

At  $t=0$ :

Index price = USD 88

Option's strike price = USD 85

At  $t=1$ , we have the up move and the down move:

For the up move, the index price is USD  $88 \times e^{0.21} = \text{USD } 108.56$ ; the option's strike price is USD 85 and the option value is therefore USD  $108.56 - \text{USD } 85 = \text{USD } 23.56$

For the down move, the index price is USD  $88 \times e^{-0.21} = \text{USD } 71.33$ . Given that the index price (USD 71.33) is below the strike price (USD 85), the option is valueless (value of zero).

Finally, we need to weight both values to get the price of the option at  $t=0$ :

Price at  $T_0 = (23.56 \times 0.5186 + 0 \times (1 - 0.5186)) \times e^{-3.0\%} = 11.86$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 14: Binomial Trees,**  
**Learning objectives:**

- **Calculate the value of an American and a European call or put option using a one-step and two-step binomial model.**
- **Describe how volatility is captured in the binomial model**

86. A Treasury department analyst calculates that the open EUR currency position of Garden Bank is EUR 1,000,000. The Euro is currently worth USD 1.15, and the exchange rate has the annual volatility of 25%. The risk-free rates on USD and EUR are 2% and 0.5%, respectively. To hedge the position, the analyst suggests selling a 3-month call option with a strike price of 1.1.

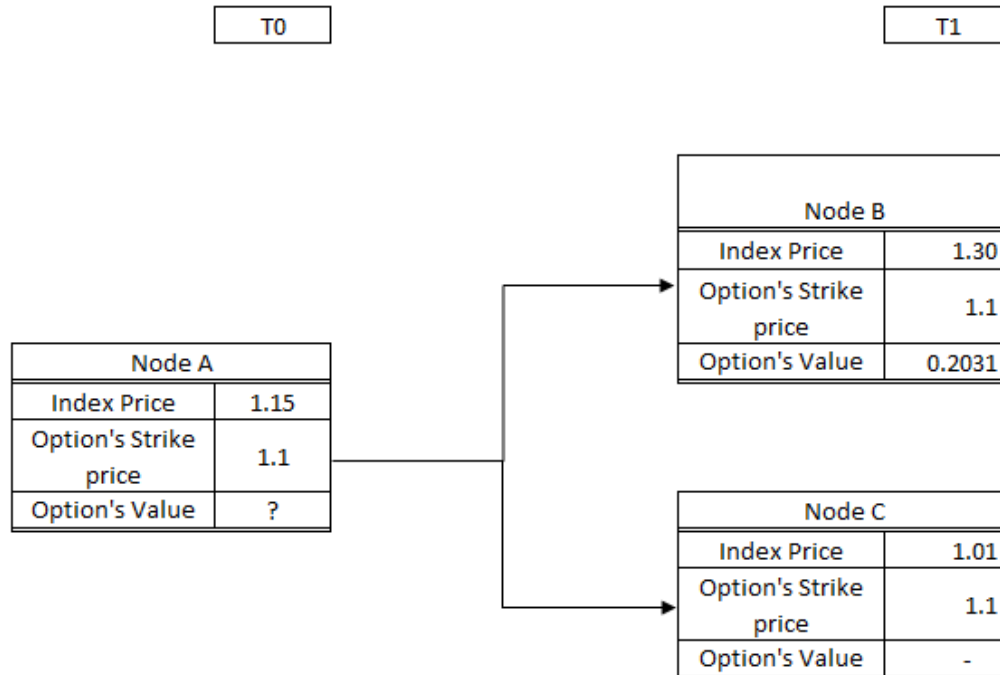
What is the price of the European call option using a one-step binomial tree model?

- A. USD 0.0958
- B. USD 0.0978
- C. USD 0.0988
- D. USD 0.0998

Correct answer: B

Explanation: 
$$p_{up} = \frac{e^{(r_{USD} - r_{EUR}) \times \Delta t} - d}{u - d} = \frac{e^{(2\% - 0.5\%) \times 0.25} - e^{-0.25 \times \sqrt{0.25}}}{e^{0.25 \times \sqrt{0.25}} - e^{-0.25 \times \sqrt{0.25}}} = 0.4838,$$

The figure below shows the stock price and the corresponding option's intrinsic value at each node.



Price at Node A =  $(0.2031 \times 0.4838 + 0 \times (1 - 0.4838))e^{-2.0\% \times \frac{3}{12}} = 0.0978$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 14: Binomial Trees, Learning objective:**  
**Calculate the value of an American and a European call or put option using a one-step and two-step binomial model.**

87. Alina Mue is analyzing the performance of ABS stocks. Stocks are currently traded on the market at USD 120 and have an expected return of 25% per year, with the annual volatility of returns of 60%. Mue is requested to calculate a 95% confidence interval for the stock price in three months under the assumption of the lognormal distribution of returns.

Which of the following is closest to the confidence interval calculated by Mue?

- A. (USD 92.02; USD 150.61)
- B. (USD 67.83; USD 219.86)
- C. (USD 102.37; USD 145.68)
- D. (USD 60.70; USD 245.67)

Correct answer: B

Explanation: Let's denote  $S_T$  – the stock price at T.

$$\ln S_T \sim \phi \left( \ln S_0 + \left( \ln S_0 - \frac{\sigma^2}{2} \right) \times T, \sigma^2 \times T \right)$$

$$\ln S_T \sim \phi \left( \ln 120 + \left( 25\% - \frac{60\%^2}{2} \right) \times 0.25, 60\%^2 \times 0.25 \right)$$

$$\ln S_T \sim \phi (4.8050, 0.0900)$$

There is a 95% probability that the normally distributed variable has a value within 1.96 standard deviations of its mean. We now have to use the standard deviation  $\sqrt{0.0900} = 0.3$  to get the confidence interval.

$$4.8050 - 1.96 \times 0.3 < \ln S_T < 4.8050 + 1.96 \times 0.3$$

$$e^{4.8050 - 1.96 \times 0.3} < S_T < e^{4.8050 + 1.96 \times 0.3}$$

$$67.83 < S_T < 219.86$$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 15: The Black-Scholes-Merton Model,**  
**Learning objective: Explain the lognormal property of stock prices, the distribution of rates of return, and the calculation of expected return.**

88. Consider a European call option with the following parameters:

|                           |          |
|---------------------------|----------|
| <b>Strike price</b>       | USD 48   |
| <b>Expiration</b>         | 6 months |
| <b>Underlying's Price</b> | USD 50   |
| <b>Annual volatility</b>  | 25%      |

Assuming a risk-free annual rate of 8%, what is the probability that the option will be exercised in a risk-neutral world? (If required, use the table at the beginning of the document for statistical calculations.)

- A. 0.70
- B. 0.30
- C. 0.64
- D. 0.36

Correct answer: C

Explanation:

Black-Scholes- Merton formulas for the prices of European call and put options:

$$C = S_0 N(d_1) - K e^{(rT)} N(d_2)$$

The term  $N(d_2)$  in the equation is the probability that a call option will be exercised in a risk-neutral world. (Note: The  $N(d_1)$  term is not quite so easy to interpret. The expression  $S_0 N(d_1) e^{rT}$  is the expected stock price at time T in a risk-neutral world when stock prices less than the strike price are counted as zero.) The strike price is only paid if the stock price is greater than  $K$  and this has a probability of  $N(d_2)$ . Therefore:

Probability of exercising the option =  $N(d_2)$

$$d_2 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{\sigma^2}{2}\right) \times T}{\sigma \times \sqrt{T}} = \frac{\ln\left(\frac{50}{48}\right) + \left(8\% - \frac{25\%^2}{2}\right) \times 0.5}{25\% \times \sqrt{0.5}} = 0.36$$

$$N(0.36) = 0.64$$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 15: The Black-Scholes-Merton Model,**  
**Learning objective: Compute the value of a European option using the Black-Scholes-Merton model on a non-dividend-paying stock.**

89. The risk manager of a large investment bank is reviewing the bank's investments in options contracts. He is particularly interested in call options contracts on shares of Hamilton Invest that the bank bought a few months ago. Hamilton Invest just unexpectedly announced that they would pay a USD 3 dividend per share in the sixth and twelfth months. The risk manager is concerned with the impact of dividends on the option's price. The risk-free rate is 5%, and the option has the following characteristics:

|                           |           |
|---------------------------|-----------|
| <b>Strike price</b>       | USD 140   |
| <b>Expiration</b>         | 13 months |
| <b>Underlying's price</b> | USD 151   |
| <b>Annual volatility</b>  | 35%       |

By how much will the price of the option change after the announcement of the dividends? Assume that  $N(d_1)$  before and after the announcement of the dividend is 0.7654 and  $N(d_2)$  before and after the announcement of the dividend is 0.5489?

- A. The price of the option will increase by USD 3.32
- B. The price of the option will decrease by USD 3.32
- C. The price of the option will decrease by USD 3.82
- D. The price of the option will decrease by USD 4.42

Correct answer: D

Explanation: Price of the call option before the announcement of the dividend:

$$c_{\text{before}} = S_0 \times N(d_1) - K \times e^{-t \times T} \times N(d_2)$$

Price of the call option after the announcement of the dividend:

$$c_{\text{after}} = (S_0 - \text{PV}(\text{dividends})) \times N(d_1) - K \times e^{-t \times T} \times N(d_2)$$

Difference:

$$c_{\text{after}} - c_{\text{before}} = -\text{PV}(\text{dividends}) \times N(d_1) = -(3 \times e^{-5\% \times 0.5} + 3 \times e^{-5\% \times 1}) \times 0.7654 = -4.42$$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 15: The Black-Scholes-Merton Model, Learning objectives:**

- **Compute the value of a European option using the Black-Scholes-Merton model on a non-dividend-paying stock.**
- **Compute the value of a European option using the Black-Scholes-Merton model on a dividend-paying stock, futures and exchange rates**

90. A trader at an investment bank was requested to value non-standardized options on stocks of RNK Construction using the Black-Scholes-Merton model. Stocks are currently trading at USD 120 per share on the market, and the risk-free rate is 5%. The options contract has the following features:

|                     |              |
|---------------------|--------------|
| <b>Option type</b>  | European Put |
| <b>Strike price</b> | USD 100      |
| <b>Expiration</b>   | 1 year       |

The trader calculates the following parameters for the model:

$N(d_1) = 0.7337$ ; and  $N(d_2) = 0.6313$ .

The price of the option is closest to:

- A. USD 2.88
- B. USD 3.01
- C. USD 3.12
- D. USD 3.47



Correct answer: C

Explanation:  $p = -120 \times (1-0.7337) + 100 \times e^{-5\%} \times (1-0.6313) = 3.12$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 5: The Black-Scholes-Merton Model, Learning objective: Compute the value of a European option using the Black-Scholes-Merton model on a non-dividend-paying stock.**

91. Consider a call option on a non-dividend paying stock where the stock price is USD 79, the risk-free rate is 5%, the time to maturity is 1 month, and  $N' = 0.3788$ . By how much will a 1% increase in the volatility of the underlying change, the value of the call option?

- A. -USD 0.0249
- B. -0.0863
- C. +USD 0.0249
- D. +USD 0.0863

Correct answer: D

Explanation: The Vega of the call option =  $S_0 \times \sqrt{T} \times N'(d_1) = \text{USD } 79 \times \sqrt{(1/12)} \times 0.3788 = 8.6387$

Thus, a 1% (0.01) increase in the volatility increases the value of the option by approximately  $0.01 \times 8.6387 = 0.0863$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 16: Option Sensitivity Measures: The Greek Letters, Learning objective: Define and describe theta, gamma, vega, and rho for option positions and calculate the gamma and vega for a portfolio.**

92. The Treasury Department of a regional bank recently got additional liquidity, amounting to USD 10,000,000. Gayane Gag, head of the treasury department, is interested in a T-Bond that is maturing at the end of September 2017. The tranche is paying a coupon rate of 2.5%. Although the tranche is not actively traded on the market, Gayane manages to collect the following information:

|                  | <b>Maturity</b> | <b>Coupon Rate</b> | <b>Price (per USD 100 face value )</b> | <b>Coupon frequency</b> |
|------------------|-----------------|--------------------|----------------------------------------|-------------------------|
| <b>Tranche 1</b> | 30/09/2018      | 1.5%               | USD 97.54                              | Semiannual              |
| <b>Tranche 2</b> | 30/09/2018      | 2.5%               | n/a                                    | Semiannual              |
| <b>Tranche 3</b> | 30/09/2018      | 3%                 | USD 103.72                             | Semiannual              |

All three tranches have a semiannual frequency of coupon payments.

Based on the above information, what is the price of Tranche 2?

- A. USD 101.527
- B. USD 101.887
- C. USD 103.028
- D. USD 101.66

Correct answer: D

Explanation: Cash flow at maturity:

|                  | <b>Notional</b> | <b>Coupon</b> |
|------------------|-----------------|---------------|
| <b>Tranche 1</b> | 100             | 0.75          |
| <b>Tranche 2</b> | 100             | 1.25          |
| <b>Tranche 3</b> | 100             | 1.50          |

Based on the law of one price, if we can construct a portfolio (using tranches 1 and 3) that has the same cash flow as Tranche 2, then both should have the same price.

Suppose we invest  $x\%$  in Tranche 1 and  $(100-x)\%$  in Tranche 3. In order to ensure the same cash flow  $x$ , the formula is the following:

$$x \times 0.75 + (1-x) \times 1.5 = 1.25 \rightarrow x = 33.33\%$$

33.33% invested in Tranche 1 and 66.67% invested in Tranche 3 will cost:

$$33.33\% \times \text{USD } 97.54 + 66.67\% \times \text{USD } 103.72 = \text{USD } 101.66$$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 9: Pricing Conventions, Discounting, and Arbitrage, Learning objective: Construct a replicating portfolio using multiple fixed income securities to match the cash flows of a given fixed-income security.**

93. A risk manager prepares a presentation on the interest rate risk of the bank's bond portfolio. The table below shows the value of the portfolio in cases of key rate shifts and corresponding key rate '01s.

|               | Value   | Key Rate '01 |
|---------------|---------|--------------|
| Initial Curve | 900.525 |              |
| 2-year Shift  | 898.950 | 1.5750       |
| 5-year Shift  | 900.302 | 0.2230       |
| 10-year Shift | 889.112 | 11.4130      |
| 30-year Shift | ?       | -0.5980      |
| Total         |         | 12.613       |

What are the portfolio's value and key rate duration for a 30-year shift?

- A. Portfolio's value = 901.123; Key rate duration = -6.641
- B. Portfolio's value = 901.123; Key rate duration = -6.641
- C. Portfolio's value = 899.927; Key rate duration = 135.090
- D. Portfolio's value = 899.927; Key rate duration = -9.119

Correct answer: A

Explanation: Portfolio value after a 30 year shift =  $900.525 - (-0.598) = 901.123$

$$\begin{aligned}
 \text{KR duration for a 30 year shift} &= \frac{10,000 \times (01 \text{ Measure})}{\text{Value of the Portfolio}} \\
 &= \frac{10,000 \times (-0.5980)}{900.525} \\
 &= -6.641
 \end{aligned}$$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 14: Modeling and Hedging Non-Parallel Term Structure Shifts, Learning objectives:**

- *Describe key-rate shift analysis.*
- *Define, calculate, and interpret key rate '01 and key rate duration.*
- *Describe the key rate exposure technique in multi-factor hedging applications; summarize its advantages and disadvantages.*
- *Calculate the key rate exposures for a given security and compute the appropriate hedging positions given a specific key rate exposure profile*

94. A generalized autoregressive conditional-heteroskedastic (GARCH)(1,1) model has the following parameters:

$$\omega = 0.0005; \alpha = 0.01; \beta = 0.98$$

The implied long-run volatility level is closest to:

- A. 0.05
- B. 0.2236
- C. 0.0003
- D. 0.0159

Correct answer: B

Explanation: The long-run average variance =  $\omega / (1 - \alpha - \beta) = 0.0005 / (1 - 0.01 - 0.98) = 0.05$

The long-run volatility =  $\sqrt{\omega / (1 - \alpha - \beta)} = \sqrt{0.05} = 0.2236$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 3: Measuring and Monitoring Volatility, Learning objective: Apply the exponentially weighted moving average (EWMA) approach and the GARCH (1,1) model to estimate volatility**

95. A risk analyst uses the EWMA model with  $\lambda = 0.85$  to carry out an update of correlation and covariance rates. On day  $n - 1$ , the observed percentage changes for variables  $X$  and  $Y$  are 5% and 3%, respectively. Historical data puts the correlation estimate between  $X$  and  $Y$  at 0.66 on day  $n - 1$ . If the estimated standard deviations on day  $n - 1$  are 2.9% and 3.5% for  $X$  and  $Y$ , respectively, then which of the following is closest to the new estimate of the covariance between  $X$  and  $Y$  on day  $n$ ?

- A. 0.5612
- B. 0.0008
- C. 0.0011
- D. 0.0014

Correct answer: B

Explanation:  $\text{Cov}_n = \lambda \times \text{cov}_{n-1} + (1 - \lambda)X_{n-1} \times Y_{n-1}$

But first, we must determine  $\text{Cov}_{n-1}$

We know that  $\text{Cov}_{xy} = \rho_{xy} \times \sigma_x \sigma_y$

$$\text{Cov}_{n-1} = 0.66 \times 0.029 \times 0.035 = 0.0006699$$

$$\text{And } \text{Cov}_n = 0.85 \times 0.0006699 + 0.15 \times 0.05 \times 0.03 = 0.0008$$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 3: Measuring and Monitoring Volatility, Learning objective: Apply the exponentially weighted moving average (EWMA) approach and the GARCH (1,1) model to estimate volatility**

96. Almytra Bank approved a loan with a total commitment of USD 108 million. The bank disbursed an amount of USD 66 million, which is also the total outstanding loan amount. The credit risk department provided the following details with regard to the riskiness of the loan:

- The standard deviation of the probability of default and the loss rates are 3% and 11%, respectively
- The probability of default over the next year is 1.5%
- The loss rate if the customer defaults is 22 percent

What is the unexpected loss for the customer?

- A. USD 990,131.99
- B. USD 449,006.20
- C. USD 544,500
- D. USD 437,233.50

Correct answer: A

Explanation: Unexpected loss = Exposure amount  $\times \sqrt{\{\text{Probability of default} \times (\text{Standard deviation of loss rate})^2 + \text{Loss rate}^2 \times \text{Standard deviation of the probability of default}^2\}}$

Unexpected loss = USD 66 million  $\times \sqrt{\{0.015 \times (0.11)^2 + 0.22^2 \times 0.03^2\}} = \text{USD } 990,131.99$

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 6: Measuring Credit Risk, Learning objective: Define and explain unexpected loss (UL).**

97. Austral-Asian Bank uses the advanced measurement approach to calculate the operational risk capital for regulatory purposes. Samuel Baker, a financial controller, working at the bank, analyzes the research reports on internal data and external data. He concludes the following from his analysis:

- I. The loss frequency distribution must be specific to the bank and based on internal data and scenario analysis estimates
- II. Historically, banks have done a much better job at documenting their operational losses than their credit risk losses

Which of these statements are incorrect?

- A. I only
- B. II only
- C. I and II
- D. None of the above

Correct answer: B

Explanation: Statement I is correct. The loss frequency distribution must be specific to the bank and based on internal data and scenario analysis estimates.

Statement II is incorrect. Banks have done a much better job at documenting their credit risk losses than their operational losses.

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 7: Operational Risk, Learning objectives:**

- ***Explain how a loss distribution is derived from an appropriate loss frequency distribution and loss severity distribution using Monte Carlo simulations.***
- ***Describe how to use scenario analysis in instances when data is scarce.***

98. An investment bank has hired external consultants to review its stress testing procedures. After a month of thought investigations, the consultants came up with the following comments:

I. The bank has ensured an appropriate coverage in stress-testing activities, as the stress testing is applied to the business line, portfolio, and risk type levels, and not on an enterprise-wide basis.

II. Stress testing should be conducted over various relevant time horizons. The bank should concentrate not on conditions that may materialize in the near term but also on adverse situations that take longer to develop.

Which of the comments is (are) correct?

- A. Comment I only
- B. Comment II only
- C. Neither of the comments is correct
- D. Both comments are correct

Correct answer: B

Explanation: Comment I is incorrect. An effective stress testing framework should be applied at various levels in the banking organization, such as business line, portfolio, and risk type, as well as on an enterprise-wide basis.

Comment II is correct. Stress testing should be conducted over various relevant time horizons. The bank should concentrate not on conditions that may materialize in the near term but also on adverse situations that take longer to develop.

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 8: Stress Testing, Learning objectives:**

- ***Describe the rationale for the use of stress testing as a risk management tool.***
- ***Identify key aspects of stress testing governance, including choice of scenarios, regulatory specifications, model building, stress-testing coverage, capital, and liquidity stress testing and reverse stress testing.***
- ***Describe the relationship between stress testing and other risk measures, particularly in enterprise-wide stress testing.***

99. A risk manager wants to stress a bank's USD 320 million real-estate loan portfolio. Although at the moment the average probability of default (PD) for the portfolio is pretty high (11.5%), the bank's management doesn't worry about the improvement of the credit quality of the portfolio as the collateral for the loans is a very expensive commercial property that always had high demand.

Based on the experience of 2008, the risk manager assumes a sequence of similar negative economic events that would not only increase the PD to 20% of the portfolio but also drastically decrease recovery rates on the loans to 40%.

What is the real-estate portfolio's expected loss in the stressed scenarios?

- A. USD 10,800,000
- B. USD 14,720,000.
- C. USD 25,600,000.
- D. USD 38,400,000.

Correct answer: D

Explanation: Stressed scenario:

Expected loss = PD × LGD × EAD

LGD = (1 – Recovery rate) = (1 – 40%)

Expected loss = 20% × (1 – 40%) × USD 320,000,000 = USD 38,400,000

**FRM Part 1, Book 4: Valuation and Risk Models, Chapter 8: Stress Testing, Learning objective: Identify key aspects of stress testing governance, including choice of scenarios, regulatory specifications, model building, stress-testing coverage, capital, and liquidity stress testing and reverse stress testing.**

100. Which of the following statements regarding stress testing methodologies is incorrect?

- A. The impact of stress tests is evaluated against the economic profit or loss
- B. An appropriate stress testing framework should comprise a broad range of scenarios covering risks at different levels of granularity
- C. The stress testing program should explicitly cover complex and bespoke products such as securitized exposures
- D. A bank should enhance its stress testing methodologies to capture the effect of reputational risk

Correct answer: A

Explanation: The impact of stress tests is usually evaluated against one or more measures. Typically used measures are asset values, accounting, and economic profit and loss, regulatory and economic capital, and liquidity and funding gaps.

***FRM Part 1, Book 4: Valuation and Risk Models, Chapter 8: Stress Testing, Learning objectives:***

- *Describe the rationale for the use of stress testing as a risk management tool.*
- *Identify key aspects of stress testing governance, including choice of scenarios, regulatory specifications, model building, stress-testing coverage, capital, and liquidity stress testing and reverse stress testing.*
- *Describe the relationship between stress testing and other risk measures, particularly in enterprise-wide stress testing.*